

Techno-economic analysis of a PV/WT/biomass off-grid hybrid power system for rural electrification in northern Morocco using HOMER

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ABSTRACT

The increasing environmental concerns associated with fossil fuels have elevated the significance of sustainable energy sources such as solar, wind, and biomass. This study aims to design a hybrid renewable energy system capable of meeting the annual energy demand of residential areas in Zoumi's circle, estimated at 15545.13 kWh/day. Using HOMER Pro software, we analyze various combinations of renewable resources (solar, wind, biomass) and storage solutions (batteries) to assess their electricity generation, cost implications, and environmental impact. The study identifies an optimal hybrid configuration that includes solar, wind, biomass, batteries, and converters. This integrated system proves to be the most cost-effective option, achieving an energy cost of 0.125 \$/kWh and a net present cost (NPC) of 8.29 \$ million. It generates 11.14 GWh/year and achieves a 100 % renewable energy ratio, significantly reducing CO₂ emissions by approximately 5900 tons/year compared to diesel generators.

1. Introduction

The increasing global demand for energy poses significant challenges to sustainability, economic development, and environmental conservation. Factors such as population growth, urbanization, and industrialization drive this demand, exacerbated by technological advancements and evolving lifestyles [1]. In Morocco, these trends are underscored by varied electricity tariffs that reflect consumption levels and regulatory changes aimed at managing energy costs. Electricity prices vary based on consumption levels, starting at 0.9 MAD/kWh for usage up to 100 kWh per month and reaching 1.60 MAD/kWh for consumption surpassing 500 kWh per month [2]. The resulting dynamic between energy demand and supply affects both the cost and sustainability of energy sources, impacting policymakers, industry stakeholders, and consumers alike. Transitioning to sustainable energy solutions is crucial for long-term environmental and economic stability, particularly in rural areas facing electricity shortages during peak demand periods.

Zoumi's circle, located in northern Morocco, experiences frequent electricity cuts, especially during summer months. This is primarily due to the increased demand from farmers using water pumps to irrigate their farms, highlighting the critical need for reliable and sustainable energy solutions tailored to local conditions. Hybrid renewable energy

systems, integrating sources such as solar, wind, and storage technologies, offer a promising solution to mitigate energy costs and environmental impacts while enhancing energy resilience in communities like Zoumi's circle [3].

This study focuses on evaluating the feasibility of a hybrid solar-wind energy system to meet the specific energy demands of Zoumi's circle. By assessing technical feasibility, economic viability, and policy implications, the research aims to optimize system configurations and support sustainable energy adoption in rural Morocco. The research gap addressed by this study lies in the application of hybrid renewable energy systems in northern Morocco, particularly in communities facing electricity shortages due to agricultural demand. While existing literature provides foundational knowledge (Table 1), further exploration is needed to understand how these systems can be effectively implemented in local socio-economic and geographic contexts.

Designing the optimal configuration for Renewable Energy Systems (RES) involves considering various factors, including geographical location, energy consumption patterns, and the compatibility of energy sources. Numerous research studies have explored the incorporation of diverse sustainable energy sources in hybrid systems to fulfill energy requirements, presenting several advantages such as enhancing the availability of renewable energy, improving energy reliability in rural

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Table 1

Techno-economic studies on integrating renewable resources with other energy systems.

Study	Year	Description	Study objective
Araoy et al. [5]	2024	The study investigates the techno-economic modeling and optimal sizing of autonomous hybrid microgrid renewable energy systems for sustainable rural electrification in the Nsukka Community, comprising 88 villages.	It aims to develop a size optimization model using the Grasshopper Optimization Algorithm (GOA) to minimize the Total Net Present Cost (TNPC), unmet load, and Cost of Energy (COE).
Hassan et al. [6]	2024	This study evaluates the potential for solar and biomass energy development across various regions in Iraq using a geographic information system (GIS) analysis.	It aims to assess strategic zoning for biomass energy and analyze the energy surplus factor, which indicates a region's ability to produce more energy than it consumes. The study identifies disparities in energy production-consumption balances, segments regions into tiers based on their surplus factors, and provides insights for optimizing energy production, managing consumption, and reducing CO ₂ emissions.
Susilowati et al. [7]	2023	This study evaluates the economic feasibility and carbon credit potential of a hybrid biomass-solar PV-battery power plant for electrifying remote areas in Indonesia, specifically in Sakti, Batukandik, and Batumadeg on Nusa Penida, Bali.	It aims to determine the project's economic viability through internal rate of return (IRR) and net present value (NPV) analysis, factoring in carbon credits and biochar sales, and to assess the project's impact on job creation and its contribution to sustainable development goals and national economic activities in Indonesia.
Al afif et al. [8]	2023	This research investigates the techno-economic feasibility of off-grid and on-grid hybrid renewable energy systems for Al-Karak governorate in Jordan.	It aims to optimize renewable energy integration using HOMER Pro software by evaluating different configurations of wind turbines, biogas plants, photovoltaic panels, flywheels, and batteries to minimize net present cost, leveled cost of energy, and CO ₂ emissions.
Baş et al. [9]	2023	This study evaluates the environmental and economic benefits of integrating biogas and solar power systems into 456 urban wastewater treatment plants (WWTPs) in Turkey.	It aims to assess the energy intensity of WWTPs by type and treatment process and to investigate the cost-effectiveness of renewable energy source (RES) integration using energy system optimization software under various market-based scenarios.
Arif et al. [10]	2023	This study aims to optimize and assess the cost-effectiveness of standalone off-grid photovoltaic (PV) systems for generating electricity in rural areas	It aims to develop net zero energy (NZE) villages by evaluating the techno-economic feasibility of a PV-powered system using HOMER software, based on the solar irradiance and electric load of a selected location.

regions, reducing dependence on fossil fuels, promoting environmentally-friendly energy generation, and lowering long-term energy expenses. Moreover, they provide a practical solution for extending modern energy access to remote areas without the necessity for costly grid transmission and distribution infrastructure. Consequently, substantial investigations have been conducted to optimize their efficiency.

Certain studies concentrate on evaluating the viability of renewable energy in specific rural areas, achieving equilibrium in power generation, or determining system dimensions through mathematical models. The predominant focus of research endeavors is on identifying the most efficient method to size the elements of integrated energy storage systems within Renewable Energy Systems (RES) and optimizing energy management for optimal outcomes. The research literature extensively delves into diverse optimization strategies driven by the objective of either minimizing or maximizing specific economic or technical criteria. These criteria encompass aspects such as the Cost Of Energy (COE), Net Present Cost (NPC), the proportion of energy derived from sustainable systems, system degradation, fuel consumption, and greenhouse gas emissions.

For instance, Abraham Olatide Amole et al. [4] focused on providing an energy generation system based on photovoltaics (PV) and diesel generators (DG) for stand-alone applications in remote Nigerian communities. The study conducted a comparative analysis to assess the impact of control strategies, namely load following (LF), cycle charging (CC), and combined dispatch (CD). The designed energy systems: 60 kW PV, 78.69 kW PV, PV-DG-LF, PV-DG-CC, and PV-DG-CD were simulated using HOMER Pro. Technical, economic, and environmental indices were used as performance metrics. The study further demonstrated that PV-DG-LF is the optimal design with the highest renewable penetration of 80.7 % and the least annual total fuel consumption (TFC) of 6594 L. The economic results confirmed the suitability of PV-DG-LF with the lowest fuel cost (FC) and moderate cost of energy (COE) of 219934 \$ and 0.302 \$/kWh, respectively. PV-DG-LF was found to be more environmentally friendly, with the least annual CO₂ emission of 16629.652 kg, aligning with sustainable development goal (SDG) 7.

Similarly, Enock Mulenga et al. [11] investigated the techno-economic viability of a Photovoltaic-Diesel Hybrid system for rural electrification in sub-Saharan Africa, focusing on Chilubi island in the Northern Province of Zambia, a remote district without electricity access. Utilizing HOMER software, the study revealed that running diesel generators independently is economically impractical, resulting in a high Levelized Cost of Electricity (LCOE). Factors influencing this include fluctuations in diesel pump prices, elevated fuel transportation costs, and high operational and maintenance expenses. The study found that a 100 % Photovoltaic (PV) system with a battery setup yielded the most favorable LCOE.

Gildas Fosso Tajouo et al. [12] conducted a techno-economic assessment of a small-scale PV/Wind/Battery hybrid system designed for environmentally friendly off-grid electrification in rural areas in Mbouda. Employing HOMER software, the study revealed that, in comparison to fixed solar panels, a solar tracking system with dual axes demonstrated the highest power production. Additionally, it was observed that increasing the hub height significantly enhanced wind power generation and penetration. From an economic perspective, the solar tracking system with dual axes exhibited the lowest LCOE and NPC when compared to other solar tracking systems. Similarly, Pankaj Kumar et al. [13] examined a case study aimed at providing power to underserved isolated rural communities in Eastern India. The research utilized resources readily accessible in the local area, such as solar, biomass, and diesel energy, employing the HOMER tool to identify the most favorable system architecture. The findings revealed that the cost of electricity and net present cost for the most cost-effective mixed configuration (comprising 600 kW solar photovoltaic, 10 kW biomass generator, 50 kW diesel generator, 1000 kWh battery, and 200 kW bidirectional converter) were 0.222 \$/kWh and 0.922 million \$, respectively. The

system demonstrated advantages in the form of decreased emissions and enhanced economic efficiency.

Upon examining numerous recent articles, a significant gap was observed regarding discussions on Morocco's capacity to offer clean energies, particularly given its ambitious goal to meet over half of its electricity demands through renewable sources by 2030. Most studies focus on other regions, leaving a gap in understanding how these systems can be implemented effectively in Morocco's specific socio-economic and geographic context. This article breaks new ground by conducting a thorough assessment of the renewable energy potential in the Zoumi's circle of Morocco. It stands out as a rare resource emphasizing Morocco's potential to harness green energy, notably biomass, a type of energy often overlooked in the country's existing literature. The primary aim of this article is to explore the ideal configuration and conduct a technical and economic evaluation of a hybrid energy system that incorporates solar, wind, and biomass for rural electrification, with batteries serving as storage units.

This paper is structured as follows: Section 1 introduces the study's objectives, contextualizes the energy challenges in Zoumi's circle, and reviews relevant literature on renewable energy systems, emphasizing their applications and benefits. Section 2 details the methodology, including modeling approaches and economic analyses specific to the Zoumi region. Section 3 presents findings from simulations and optimization processes tailored to local conditions. Finally, Section 4 summarizes the conclusions drawn from the study, highlights policy implications, and suggests directions for future research.

2. Methodology

The aims of this research are to showcase the economic and technological viability of an off-grid hybrid energy system that combines solar, wind, and biomass for supplying electricity to remote rural regions. To achieve this, a specific village in North Morocco is taken as a case study. The utilization of HOMER Pro, a software developed by the

National Renewable Energy Laboratory, facilitates the simulation, design, and optimization of hybrid energy systems tailored to specific locations. This tool plays a pivotal role in identifying the most favorable configurations for hybrid systems. Through the application of the HOMER tool, the hybrid power system is devised, and optimal component sizes are determined via techno-economic analysis. The process of optimal system design and the corresponding techno-economic analysis for off-grid hybrid systems are elaborated below, and you can refer to Fig. 1 for an illustration.

Numerous methods and tools are accessible for modeling energy systems, yet the HOMER software stands out as a popular choice among both researchers and industries due to its multitude of advantages. One notable feature that distinguishes HOMER is its ability to conduct sensitivity analysis, a capability lacking in several other tools. Although the iHOGA software can simulate sensitivity analysis, this function is restricted to its pro version. Conversely, many alternative tools lack the comprehensive modeling and simulation capabilities necessary for addressing the key components outlined in this paper. Additionally, certain tools such as INSEL focus solely on technical analysis, without incorporating economic analysis capabilities. Considering these aspects, HOMER Pro software emerges as the most suitable and adaptable tool for fulfilling the specific requirements of this research [14].

2.1. Selection of region and climatic conditions

This study centers on the Zoumi circle, situated in the northern region of Morocco. The geographical coordinates for this area are approximately 34° 47' 52" North latitude and 5° 20' 58" West longitude, as illustrated in Fig. 2. An abbreviated description of the research site is provided in Table 2. Zoumi encompasses a total land area of 287 km² and accommodates a population of 56824 people, living in 12445 households, as indicated by the 2014 general census of population and housing [15].

In Zoumi, the climate exhibits distinct seasonal characteristics.

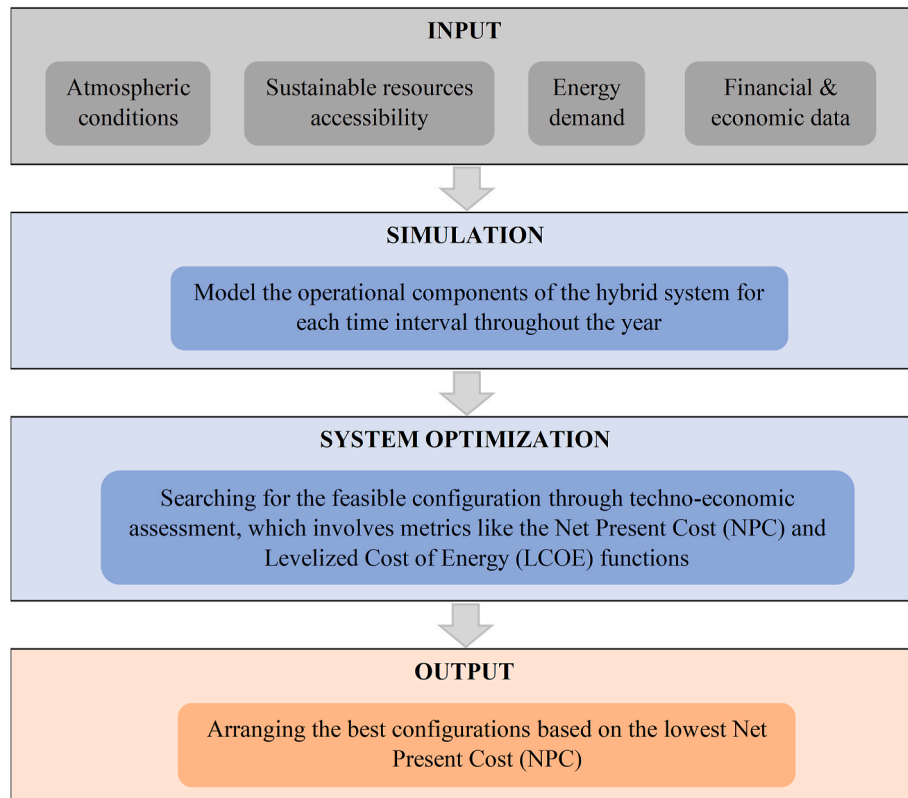


Fig. 1. Approach for optimizing the design and conducting techno-economic analysis of an off-grid hybrid power system.

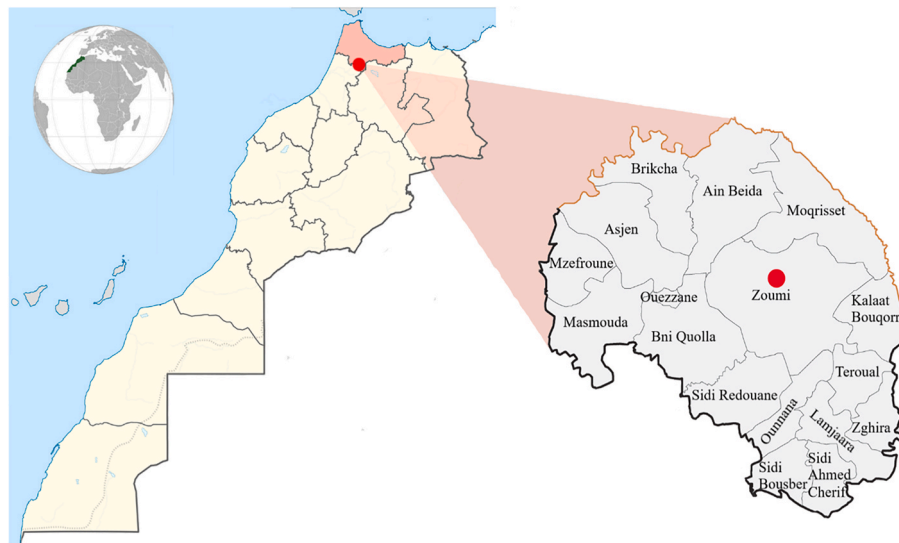


Fig. 2. Geographic location of Zoumi circle.

Table 2
General information of study area.

Province	Ouezzane
Country	Morocco
Latitude	34° 47' 52" North
Longitude	5° 20' 58" West
Altitude	369 m (1211 feet)
Total Area	287 km ²
Total population	56824
Households	12445
Population growth rate between 2004 and 2014	0.93 %

Summers are characterized by their brevity, high temperatures, arid conditions, and clear skies, while winters are prolonged, marked by cold temperatures, precipitation, and partially cloudy conditions. The temperature in this region generally varies between 4 °C and 34 °C throughout the year, with rare occurrences of temperatures falling below 0 °C or exceeding 39 °C. Table 3 offers a comprehensive summary of the region's monthly weather patterns.

The period characterized by higher levels of precipitation in Zoumi spans 7.7 months, commencing on September 27 and concluding on May 18, during which there is a greater than 16 % likelihood of any given day experiencing rainfall. Specifically, the rainy season encompasses 8.9 months, starting on September 9 and ending on June 6, with a continuous 31-day span featuring at least 13 mm of rainfall. November stands out as the month with the most substantial rainfall, averaging 104 mm. In contrast, the rain-scarce period extends for 3 months, lasting from 6 June to 9 September. Among the months, July records the lowest average rainfall, amounting to just 3 mm (See Fig. 3).

The duration of the hot season spans 2.6 months, commencing on June 22 and concluding on September 10, during which the daily high temperatures consistently surpass 30 °C. Zoumi experiences its peak heat in the month of July, registering an average high temperature of

33 °C and a low of 18 °C. Conversely, the cool season extends for 4 months, running from 17 November to 21 March, featuring daily high temperatures that remain below 18 °C. The chilliest month in Zoumi is January, with an average low of 5 °C and a high of 14 °C. For detailed temperature data, Fig. 4 (a) and 4 (b) provide visual representations of the average high and low temperatures on both a monthly and daily basis in the Zoumi circle, respectively.

The duration of daylight in Zoumi experiences notable fluctuations throughout the year. In the year 2023, the day with the least amount of daylight occurs on December 22, providing just 9 hours and 49 minutes of illumination, while the day with the most daylight falls on June 21, offering a generous 14 hours and 30 minutes of daylight. Fig. 5 illustrates the durations of daylight and twilight in Zoumi circle.

2.2. Renewable resources estimation

Fig. 6, produced using HOMER tool and utilizing data sourced from the NASA Prediction of worldwide Energy Resource (POWER) database, depicts the monthly averages for global horizontal radiation over 22-year period (July 1983–Jun 2005) in the research area. During July, the region received the most abundant solar irradiance, averaging 7.42 kWh/m² per day, while December recorded the least, with a daily average of 2.46 kWh/m².

Fig. 7 displays the average monthly wind speed at a height of 50 m above the surface of the study location throughout a 30-year period (Jan 1984–Dec 2013), as determined using HOMER Pro software. The calculated value for the mean wind speed is 3.88 m/s.

Biomass encompasses all organic materials capable of being converted into energy, with potential applications in heat or electricity generation. This study focuses on harnessing energy from livestock dung, particularly in the Zoumi circle region, where intensive livestock activity is prevalent. The livestock population, consisting of 11530 sheep, 8980 goats, 8500 cows, and 6000 mules, contributes to dung

Table 3
Zoumi's climate 2023 [16].

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average T° (°C)	9	10	12	14	18	22	26	26	22	18	13	10
Min T° (°C)	5	6	7	9	12	15	18	19	16	13	9	6
Max T° (°C)	14	15	18	20	24	29	33	33	29	24	18	15
Rainfall (mm)	88.8	81.0	66.1	59.4	34.8	8.9	3.0	3.2	18.5	63.8	103.9	103.2
Rainy days (d)	7.9	7.6	7.4	7.8	5.1	1.4	0.5	0.9	3.2	6.9	8.5	8.7
Av Sun hours (h)	10.1	10.9	12.0	13.1	14.0	14.5	14.2	13.4	12.4	11.3	10.3	9.9

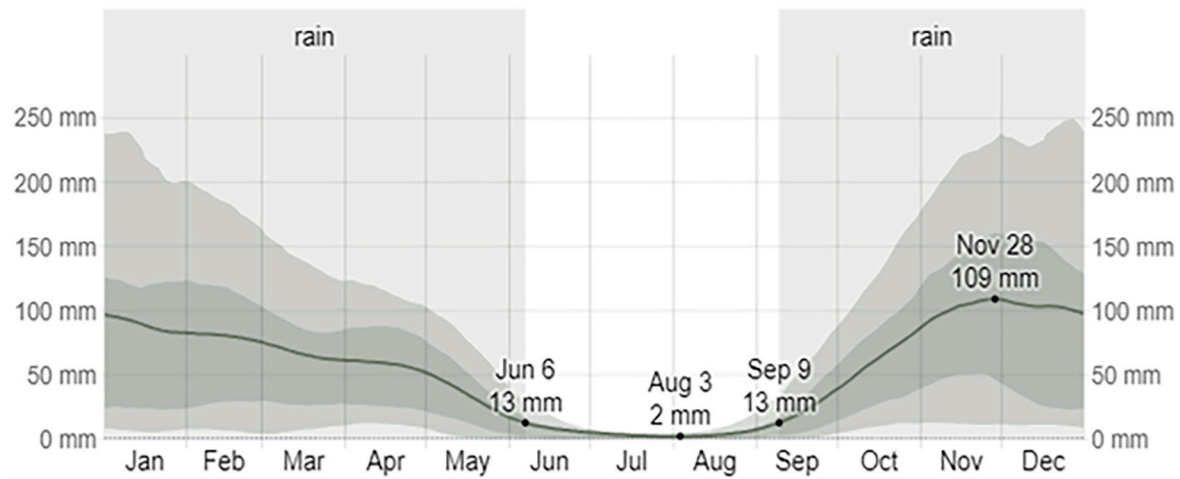


Fig. 3. Rainfall in Zoumi circle during 2023 [16].

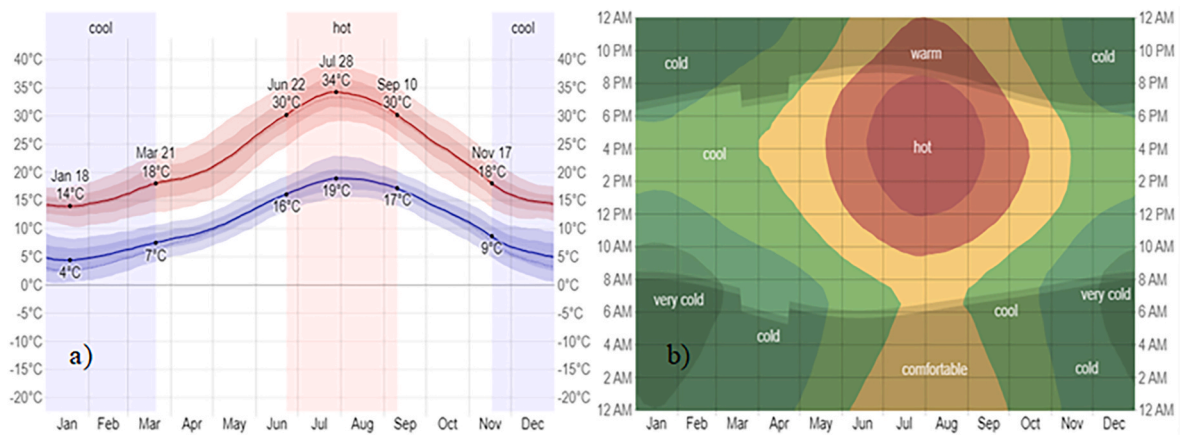


Fig. 4. The temperature in Zoumi circle during 2023. (a) The average for each month, (b) The average on a daily basis [16].

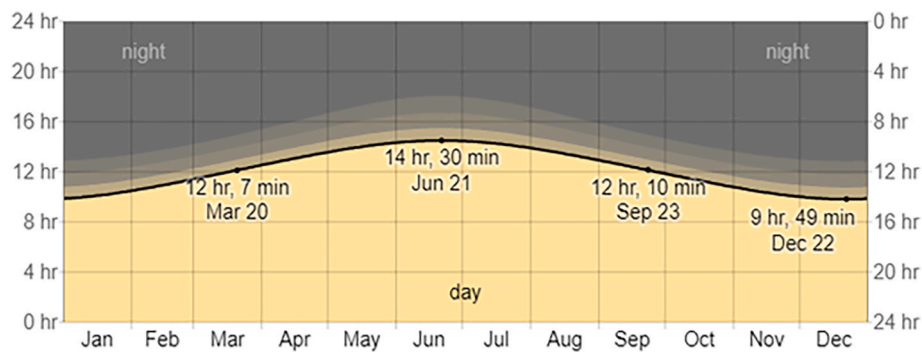


Fig. 5. Daylight and twilight in Zoumi circle during 2023 [16].

production. Assumptions include a daily dung output of 10 kg for cows and mules, and 1 kg for goats and sheep [17]. Key parameters for biogas production were considered, such as a recovery factor of 0.7, dung-to-biogas conversion rate of 0.036 m³ per kg, and biogas-to-electricity conversion of 1 kWh per 0.73 m³ of biogas [18]. The study adopts a gasification ratio of 0.05 kg/kg, a lower heating value of 5.5 MJ/kg, and assumes zero cost for biomass. The electricity potential from biomass in Zoumi circle is outlined in Table 4. The dung recovery aspect focuses on farms with provided animal feed and considers the associated collection cost as the fuel cost for this study.

Morocco stands as the sixth-largest producer of Olive Mill Wastewater (OMWW) in the Mediterranean region. Approximately 30 % of Moroccan OMWW undergoes “treatment” through natural evaporation, while over 70 % is directly discharged into the environment, posing significant environmental challenges for the country [19]. As of 2023, there were over 200 olive mills scattered across the nation, with more than 8 located in the Zoumi circle. Despite extensive research in recent years on OMWW treatment, various proposed techniques have proven inadequate, either failing to eliminate all pollutants, being economically burdensome, or generating secondary pollution necessitating additional

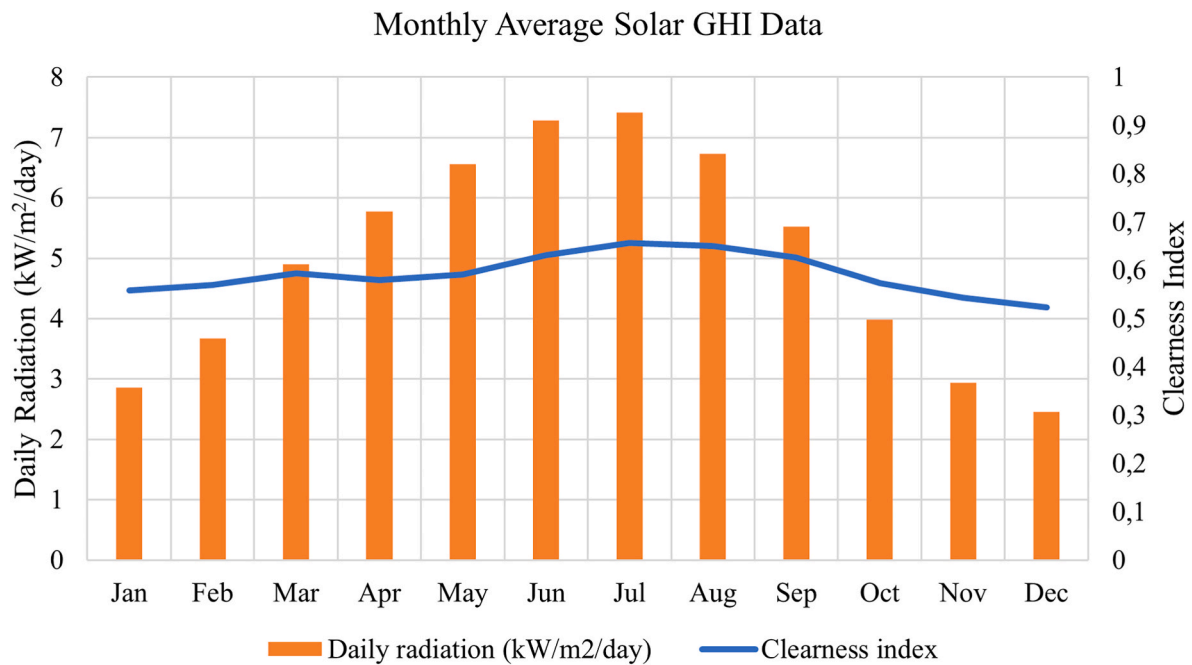


Fig. 6. Monthly solar radiation.

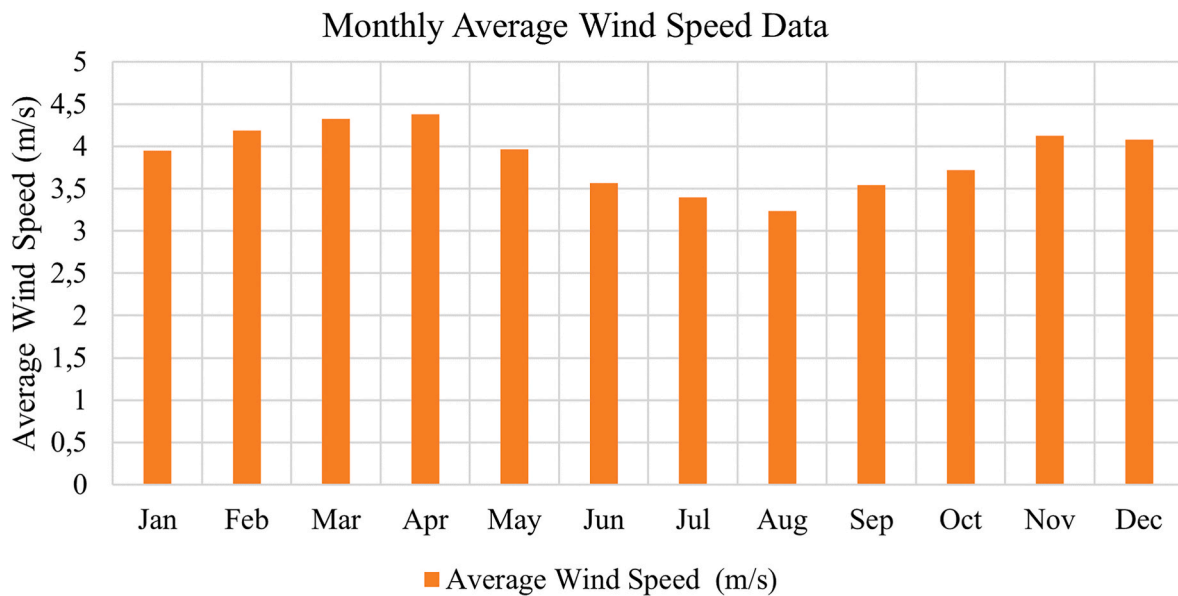


Fig. 7. Average monthly wind speeds.

Table 4
Estimation of biomass power in Zoumi circle.

Livestock	Number	Dung availability (kg/head/day)	Total dung (kg/day)	Total dung (recovery factor = 0,7)	Total biogas yield (m ³ /day)	Potential power yield (kWh/day)
Sheep	11530	1	11530	8071	290.56	398.03
Goats	8980	1	8980	6286	226.3	310
Cows	8500	10	85000	59500	2142	2934.25
Mule	6000	10	60000	42000	1512	2071.23
Total	35010		165510	115857	4170.86	5713.51

treatment. Notably, none of these techniques have achieved widespread industrial-scale implementation [20].

A promising alternative is anaerobic digestion, a biological process that transforms organic pollutants in OMWW into methane (CH₄) and

carbonic acid (H₂CO₃) [21]. This process not only decontaminates effluents with varying organic content but also produces biogas and compost, offering an environmentally sustainable solution.

In the HOMER model, the biogas generator is scheduled to operate 8

hours each day on weekdays, utilizing a daily biomass input ranging from 134 to 178 tons. Biogas derived from olive mill wastes comprises various gases and substances, including 62 % methane (CH₄), 35 % carbon dioxide (CO₂), and 2 % other components [8].

The ratio of biogas components, specifically methane, was established using Equation (1) [22]. These equations were adapted from the methodology introduced by Buswell and Mueller.

$$X_{CH_4} = \frac{(4C + H - 2O - 3N - 2S)}{8} \quad (1)$$

Where, X_{CH_4} and X_{CO_2} signify the respective proportions of methane (CH₄) and carbon dioxide (CO₂). The weight percentages (wt.%) of elemental components—C: carbon, H: hydrogen, O: oxygen, N: nitrogen, and S: sulfur.

Equation (2) [22] was employed to ascertain the quantities of CH₄ and V_M , while the overall volume of biogas, denoted as V_{bio} , was calculated based on Equation (3).

$$V_M = c \times \left(0.01 \times \frac{X_{CH_4}}{M_C} \right) \times V_I \times 0.01 VS \times 0.75 \times M_{MSW} \quad (2)$$

$$V_{bio} = \frac{V_M}{0.01 \times X_{CH_4}} \quad (3)$$

Where, V_M stands for the daily quantity of produced pure methane. (m³/day), while VS represents the amount of volatile solids in the organic waste. The coefficient for biogas production, denoted as c is considered as 0.375 g carbon/g volatile solid, assuming a 74 % deterioration of volatile solids. M_C represents the molecular weight of carbon (12 g/mol), V_I signifies the volume of one mole of gas under standard temperature and pressure (22.4 L), M_{MSW} denotes the weight of the incoming organic waste stream (kg/day), and V_{bio} indicates the daily volume of biogas produced (m³/day).

Equation (4) was utilized to estimate the generation of electrical energy from biogas. Subsequently, the electrical power generated (PE_{bio}) in Eq. (5) was determined based on the calculated electrical energy (EE_{bio}).

$$EE_{bio} = V_M \times E_C \times \eta \times 0.001 \quad (4)$$

$$PE_{bio} = \frac{EE_{bio}}{t \text{ (the working hours)}} \quad (5)$$

In this context, EE_{bio} refers to the daily output of electrical energy in kilowatt-hours (kWh), with V_M representing the daily volume of methane yield (m³/day). The methane energy constant, denoted as E_C , is considered as 10 kWh/m³. The parameter η represents the assumed electrical efficiency of the biogas engine, set at 40 %. The resulting electrical energy output (PE_{bio}) is measured in kilowatt-hours (kWh).

By employing the data and equations provided earlier, the numerical findings for the Zoumi circle have been compiled and are presented in Table 5, while Fig. 8 demonstrates the technical capacity of biomass, encompassing animal dung and olive mill waste (165.51 ton/day of dung with 134 ton/day of OMWW), within the region. Throughout the year, the accumulation of olive mill waste predominantly occurs within a concise timeframe, spanning from November to February, which will be 402 ton for each of these months.

Table 5
Availability of biomass and production yield of biogas from OMWW in Zoumi circle.

Biomass resources	OMWW potential resources
Availability (kton/year)	49–65
Availability (ton/day)	134–178
The overall biogas (m ³ /day)	32276.58–42874.86
Energy output (kWh/day)	12910,632–17149,944

2.3. Energy demand estimation

In 2022, the electrical system experienced a maximum demand of 647.72 kW, with a subsequent increase of 653.65 kW in power demand observed in 2023. Anticipating the future, Zoumi's Circle envisions a consistent average annual growth rate of approximately 0.92 % over this timeframe.

Within this investigation, the anticipated energy demand, excluding power generation, registers a lower magnitude. Nonetheless, energy consumption undergoes a gradual increase over time, influenced by diverse factors. To precisely gauge the requisite electrical energy needs, data pertinent to the region were sourced from the National Electricity Office of Zoumi's circle. Primary data pertaining to energy requisites were garnered through surveys administered within the local community, encompassing various sectors like domestic, agricultural, communal, and commercial loads. The energy demand predominantly encompasses illumination for health centers, schools, establishments, road illumination, water pumping, and minor industrial demands. The study incorporates hourly and monthly patterns of energy consumption, as illustrated in Figs. 9–11. The calculated daily aggregate energy demand reaches 15545 kWh, featuring a peak load of 2886 kW within the specified area.

2.4. Simulation and modelling of the hybrid system components

Achieving the ideal dimensions for components within a hybrid energy system PV/WT/BMG/BATT (Fig. 12) necessitates a meticulous mathematical modeling approach that encompasses both the system's components and economic considerations. This analytical process not only offers valuable insights into how the hybrid system functions under diverse scenarios but also empowers the determination of optimal sizes for individual elements. Furthermore, it facilitates a comprehensive assessment of system performance across various scenarios, ensuring a well-informed evaluation of its efficiency. This section also provides details regarding the specifications of the various components.

2.4.1. Photovoltaic modules

The output generated by a photovoltaic array can be articulated as follows [23]:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T - T_0)] \quad (6)$$

Where P_{PV} is the PV panel's output power in kilowatts (kW), f_{PV} is the derating factor (%), Y_{PV} is the whole the rated capacity of the PV array, indicating its power output under STC (kW), G_T is the sunlight radiation incident on the PV array in the current time step (kW/m²), and $G_{T,STC}$ is the standard solar radiation value per unit area of the PV panel (1 kW/m²). Additionally, α_p , T and T_0 are the temperature coefficient of power (%/°C), the actual and standard temperatures, set at 25 °C, respectively.

In this investigation, we utilized the Canadian Solar MaxPower CS6X–325P PV model, and the technical specifications of the converter can be found in Table 6. All panels were oriented with a 0-degree azimuth, and the PV system did not incorporate a tracking system. Additionally, the PV system considered a ground reflectance of 12 %, resulting in an 88 % derating factor. These factors were taken into account to accommodate the decline in PV module efficiency caused by external elements such as dust impacting the PV cells. To determine the optimal number of PVs, we employed the HOMER Optimizer software, streamlining the design process by eliminating the need for an exhaustive search.

2.4.2. Wind power generator

The conversion of mechanical energy from the wind into electrical power is the primary function of a wind generator. The evaluation of a wind site's potential relies on the available wind power. Wind turbines

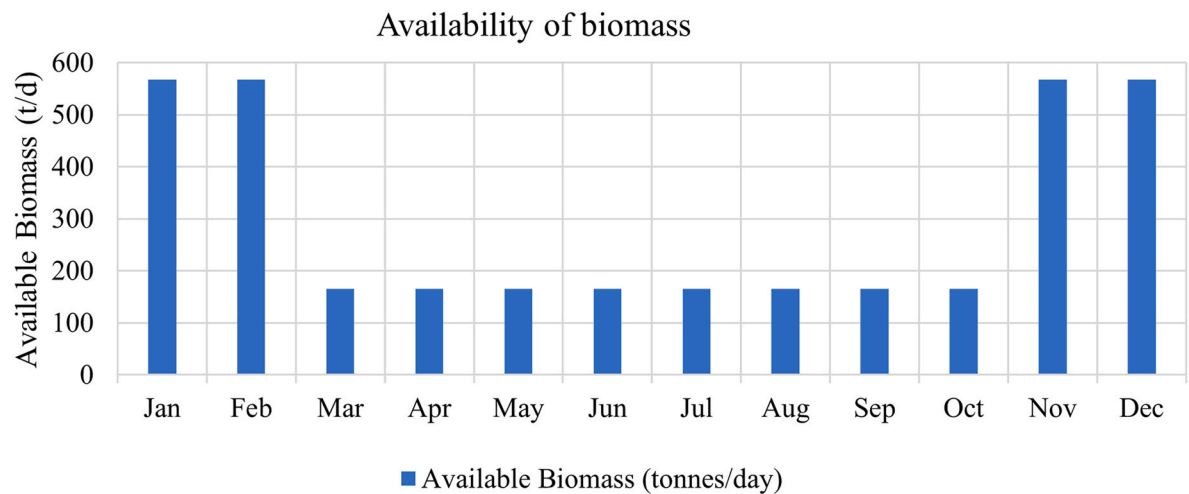


Fig. 8. Biomass technical potential for Zoumi circle, Morocco.

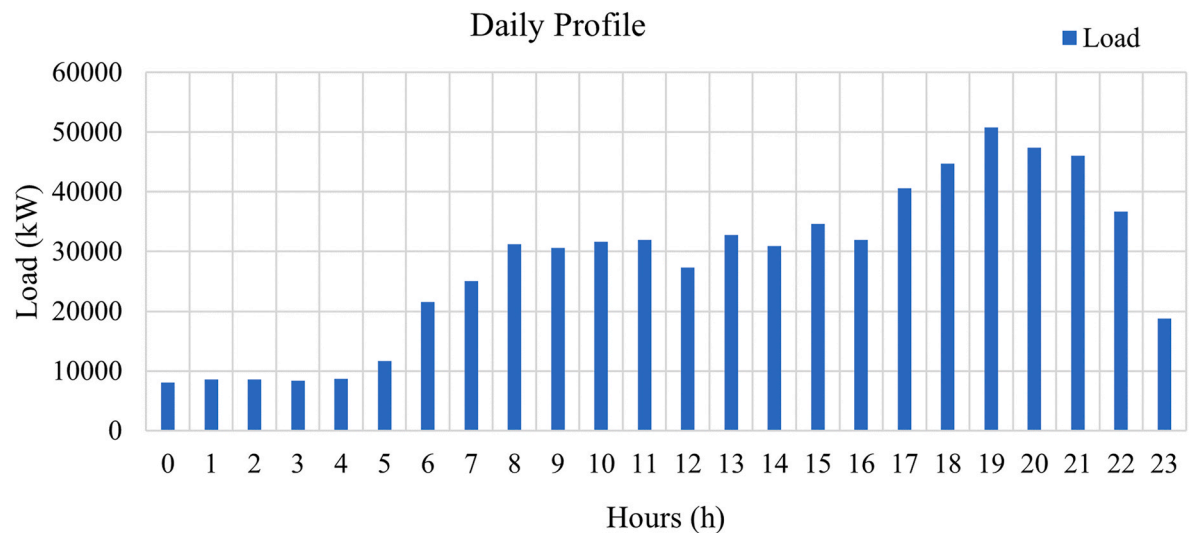


Fig. 9. The hourly load profile in Zoumi's circle.

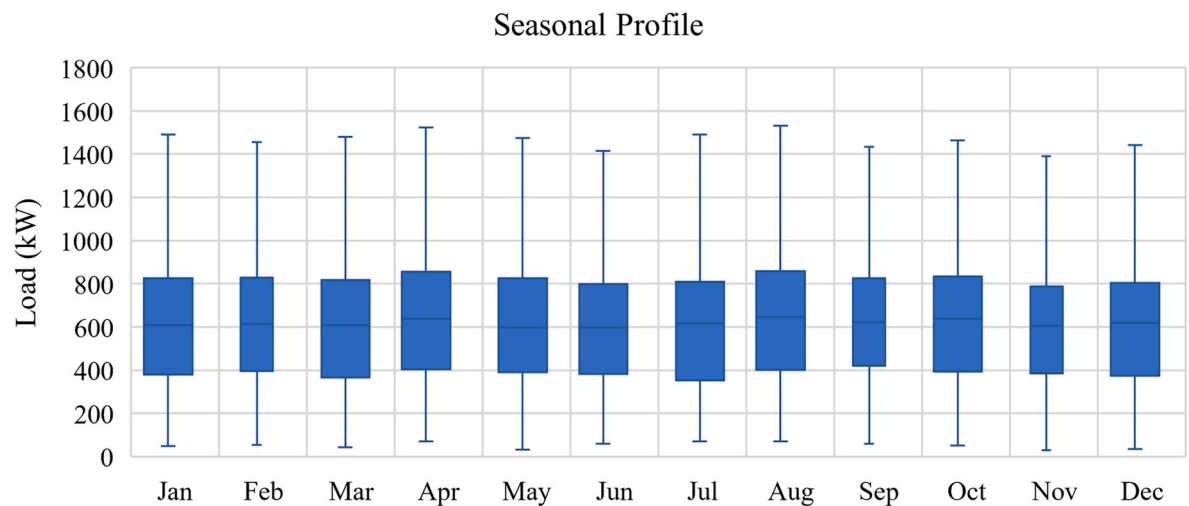


Fig. 10. The monthly load profile in Zoumi's circle.

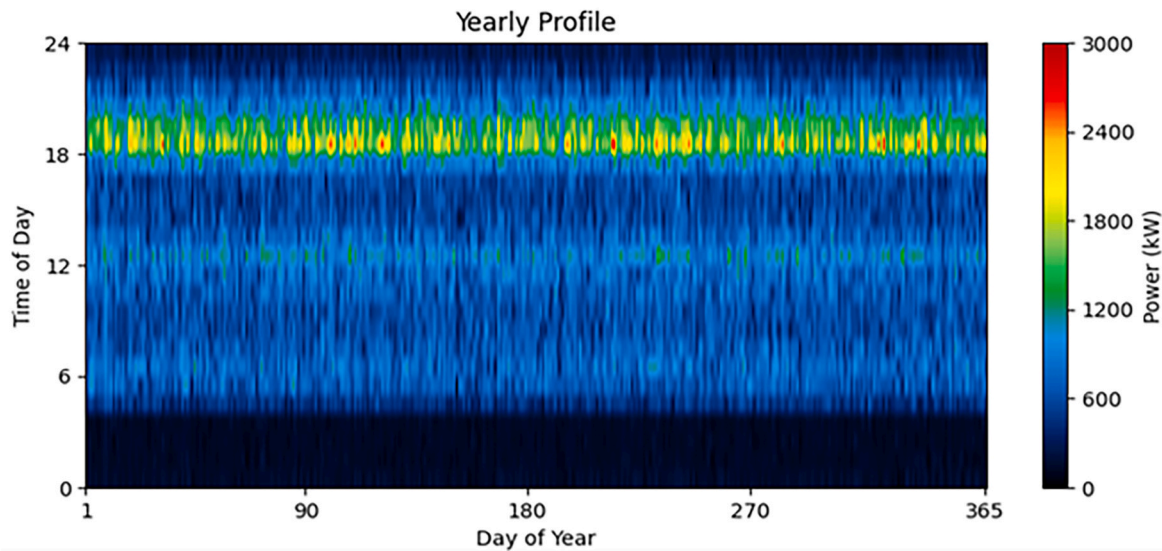


Fig. 11. The yearly load profile in Zoumi's circle.

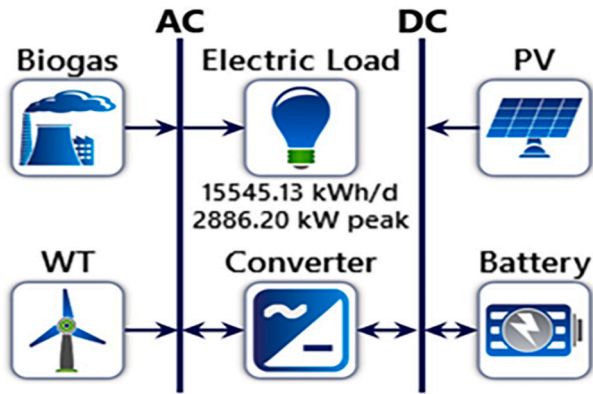


Fig. 12. Design of hybrid system in HOMER Pro.

Table 6

Technical specifications for the PV modules [24].

Type	CS6X-325P
Rated Capacity (W)	325
Optimum Operating Voltage (V)	37
Optimum Operating Current (A)	8.78
Nominal Operating Temperature (°C)	45
Temperature Coefficient (%/°C)	−0.41
Module Efficiency (%)	16.94
Lifetime (years)	25
Capital cost (\$)	185
Replacement cost (\$)	185
O&M cost (\$/year)	5.00

begin generating electricity when the wind velocity ranges from 2 to 4 m/s, initiating the rotation of the turbine blades, and continues up to 25–30 m/s, with variations based on the manufacturer and model [25]. The calculation of the output power of a wind turbine can be expressed using the following formula [26,27]:

$$P_{wt,STP(V)} = \begin{cases} P_r(V - V_{cin})/(V_r - V_{cin}) & V_{cin} \leq V \leq V_r \\ P_r & V_r \leq V \leq V_c \\ 0 & V \leq V_{cin} \text{ and } V \geq V_c \end{cases} \quad (7)$$

Where, P_r represents the rated power of the wind turbine, V_r denotes the

rated wind velocity, while V_{cin} and V_c represent the cut-in velocity and cut-out velocity, respectively.

For this research, we have chosen to analyze the Vestas V82 1.65 MW wind turbine model, produced by Vestas Wind Systems. This turbine has a rated power of 1650 kW when the wind speed is 13 m/s. The technical details of the turbine are outlined in Table 7, and its power-performance curve is depicted in Fig. 13. To determine the wind velocity at the hub height [28], the power law was employed in the calculations [29].

$$U_{hub} = U_{anem} \times \left(\frac{Z_{hub}}{Z_{anem}} \right)^\alpha \quad (8)$$

Where U_{hub} is the wind velocity at hub height of the wind generator (m/s), U_{anem} is the wind velocity at the hub height of the anemometer (m/s), Z_{hub} is Hub height (m), Z_{anem} is Anemometer height (m), and α is the exponent for power law. The rated power output is calculated using the following equation [30]:

$$P_{WTG} = \frac{1}{2} \rho A_t V^3 C_p \quad (9)$$

Where P_{WTG} is the output energy (kW), ρ is the density of air (kg/m^3), A_t is the area swept (m^2), V is the velocity of the wind (m/s), C_p is the power coefficient. In 2017, the assumed average cost of turbines per kilowatt was 925 \$ [31]. Utilizing this figure, the comprehensive costs, encompassing initial, replacement, and operation/maintenance, were computed at 1526250 \$ per unit for both initial and replacement expenditures. Additionally, an annual maintenance cost of 19078.13 \$ was factored in. For the purposes of this study, the turbine's presumed lifetime is set at 20 years.

Table 7

The chosen wind turbine model's technical specifications [32].

Model	Vestas V82 1.65 MW
Manufacturer	Vestas Wind Systems
Rated power (kW)	1650
Max hub height (m)	80
Rotor diameter (m)	82
Rated wind speed (m/s)	13
Lifetime (years)	20
Capital cost (\$)	1526250
Replacement cost (\$)	1526250
O&M cost (\$/year)	19078.13
Type	3 Blade Upwind

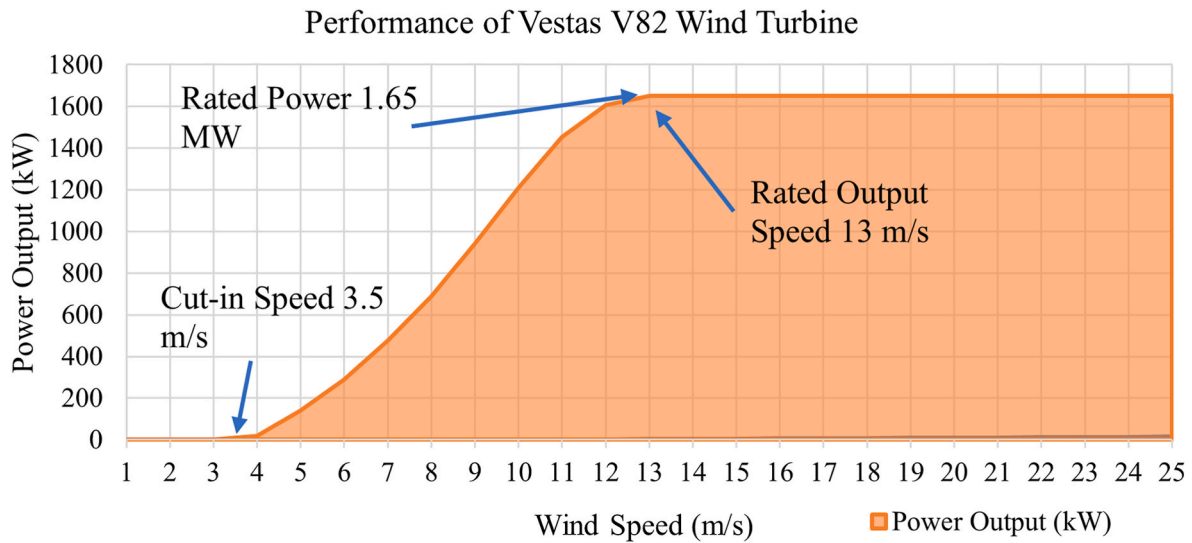


Fig. 13. The energy performance chart for the wind power generator [32].

2.4.3. Biomass generator

The energy generated per hour by the biomass system (E_{BMGS}) has been computed as follows [33]:

$$E_{BMGS}(t) = \frac{\text{Biomass availability (kg/yr)} \times CV_{BMG} \times \eta_{BMG} \times \Delta t}{365 \times 860 \times H_{BMG}} \quad (10)$$

Where the calorific value of biomass (CV_{BMG}), measured in kcal/kg, is a crucial factor in biomass energy calculations. To convert kcal to kWh, a multiplication factor of 860 is applied (since 1 kcal is equivalent to 860 kWh). The overall efficiency of the biomass generator (η_{BMG}) and the daily operational hours of the biomass generator power (H_{BMG}) play significant roles in these calculations. Additionally, Δt denotes the duration of biomass generator energy operation within a day.

The biogas generated by the biomass gasifier system serves as a versatile fuel source, offering options for powering gas turbines, gas engines, or direct combustion for bioelectricity production. This approach boasts high conversion efficiency and presents fewer pollution concerns compared to traditional fuels such as oil, natural gas, and petroleum. In this investigation, we analyze a generic 500 kW biogas genset connected to an AC output. HOMER software considers the available biogas during the generator sizing process. The characteristics of the biogas generator, along with specific parameters, are detailed in Table 8 [14,34]. Notably, the generator undergoes a scheduled two-hour maintenance shutdown every weekend from 08:00 to 10:00, as illustrated in Fig. 14.

2.4.4. Battery bank and converter system

Because of the fluctuating and sporadic characteristics of renewable resources like solar radiation, wind speed, and biomass over the course

of the day and month, enhancing the reliability of hybrid sustainable energy systems necessitates a backup mechanism, such as a battery bank. This system stores electricity generated from solar, wind, and biomass sources, supplying it during periods without sunlight, sufficient wind speed, or available biomass resources, as well as during any deficiency in energy generation. Batteries play a crucial role in maintaining a stable voltage during peak loads and efficiently distributing the stored energy [35]. In this study, A lithium-ion (Li-Ion) battery was chosen, featuring specifications including a 6 V nominal voltage, 1 kWh nominal capacity, 167 Ah maximum capacity, and a lifespan throughput of 3000 kWh per cell. For an individual battery, the considered values for both initial capital and replacement cost is 300 \$, while the annual operation and maintenance (O&M) cost is 2 \$/year for each battery [36]. The capacity of the battery in the charging state can be assessed as [37]:

$$E_{BATT}(t) = (1 - \gamma) \times E_{BATT}(t - 1) + [E_{PV}(t) + E_{WT}(t) + E_{BMGS}(t)] \times \eta_{CHA} \quad (11)$$

In this equation, $E_{BATT}(t)$ denotes the battery storage in kWh, γ is the hourly self-discharging rate of the battery, $E_{PV}(t)$, $E_{WT}(t)$, and $E_{BMGS}(t)$ represent the excess generation of the PV system, wind, and biomass generators after fulfilling the energy demand, respectively, in kWh. η_{CHA} signifies the efficiency of the battery during charging.

In the discharge phase, when the generation from renewables is insufficient to meet the energy demand, the needed power to fulfill the demand will be provided by the stored energy in the battery. The capacity of the battery during the discharging state can be assessed as [37]:

$$E_{BATT}(t) = (1 - \gamma) \times E_{BATT}(t - 1) - \frac{E_{req}(t)}{\eta_{DIS} \times \eta_{inv}} \quad (12)$$

$$E_{req}(t) = E_{DM}(t) - E_{BMGS}(t) - E_{WT}(t) - E_{PV}(t) \times \eta_{inv} \quad (13)$$

Where η_{DIS} and η_{inv} represent the discharging rate of the battery and the efficiency of the inverter, respectively. $E_{req}(t)$ is the energy needed to meet the unmet load, which is not satisfied by renewable generation, and $E_{DM}(t)$ is the total energy demand.

A crucial component in hybrid power generation systems for facilitating the connection between DC and AC electrical elements is the bidirectional power converter. This converter incorporates both a rectifier for converting DC to AC (rectification process) and an inverter for transforming AC to DC (inversion process) [38]. The selected converter for this examination comes with a lifespan of 10 years and functions with an inverter and rectifier efficiency rated at 95 %. The capital and replacement costs set at 300\$ per 1 kW converter [39],

Table 8

Characteristics and specification parameters of the biogas generator [34,35].

Model	Generic 500 kW Biogas Genset
Capacity (kW)	500
Cost of fuel (\$/kg)	0.053
Sulfur content (%)	0
Carbon content (%)	5
Density (kg/m ³)	0.720
Lower heating value (MJ/kg)	5.5
Minimum load ratio (%)	30
Lifetime (hours)	20000
Capital cost (\$)	200000
Replacement cost (\$)	200000
O&M cost (\$/op.hour)	20

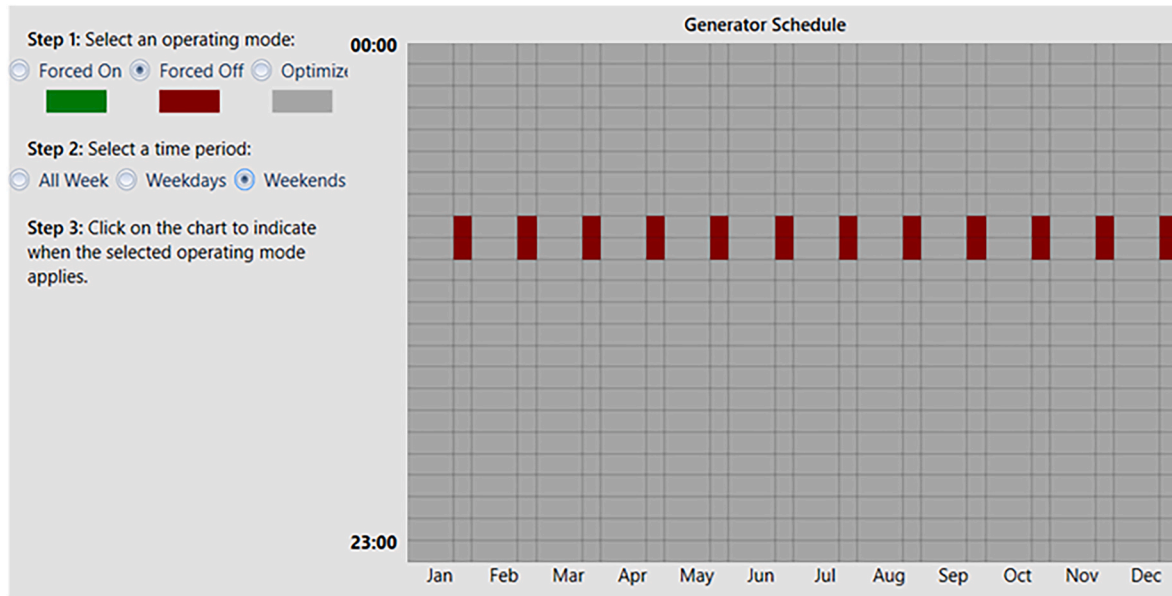


Fig. 14. The generator schedule.

Equation (14), as cited in Refs. [40,41], is utilized by the HOMER software to calculate the power output of the converter component.

$$P_{out} = P_{in} \times \eta_{inv} \quad (14)$$

2.5. Assessment criteria and sensitivity variables

In the simulation and optimization processes, the assessment of the optimization of the combined system depended on the use of NPC and LCOE as primary assessment criteria. HOMER software organizes the proposed systems based on these criteria, with a preference for lower values indicating superior system performance [42]. The evaluation criteria are outlined as follows [14,43,44]:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (15)$$

$$C_{NPC} = \frac{C_{ann}}{CRF(i, R_{prj})} \quad (16)$$

$$COE = \frac{C_{ann,tot} - C_{boiler} \times H_{thermal}}{E_{useful,tot}} \quad (17)$$

Where C_{NPC} is the net present cost, i is the real present discount rate (%) and N is number of the years, R_{prj} is the project lifespan, and CRF is the function returning the capital recovery factor, $C_{ann,tot}$ is the total annualized cost (\$/year), C_{boiler} represents the marginal cost of the boiler (\$/kWh), while $H_{thermal}$ denotes the total thermal load served (kWh/year). It is important to note that, in this context, the thermal load served in the system is assumed to be zero, thus $H_{thermal} = 0$. and $E_{useful,tot}$ is the total annual electrical load served (kWh/year).

In this section, we utilize the HOMER software analysis feature to model the economics of the combined renewable energy systems. Table 9 provides the project assumptions and input data for economic

Table 9
Overview of the project.

Name	Value	Name	Value
Project lifespan	20 years	Real present discount rate	5.73 %
Inflation rate	3 %	Maximum annual capacity shortage	0 %
Discount rate	8.90 %	Time steps per year	8760 units

parameters.

A dispatch strategy comprises a set of regulations dictating the operation of both generators and storage devices. The HOMER tool is capable of simulating two dispatch strategies: load following (LF) and cycle charging (CC) [45]. In the load following approach, a generator produces only the necessary energy to fulfill the load whenever activated. Conversely, in the cycle charging strategy, each time a generator is initiated, it operates at full capacity, and any excess energy is stored by the energy storage device. Both LF and CC strategies were taken into account in this investigation.

In the analysis of the combined system, variations in sustainable energy resources like solar irradiation, wind speed, biomass availability, and the scaled annual average electric load were incorporated as sensitivity variables in the simulations. Table 10 provides details on the sensitivity variables and their respective values.

3. Results and discussions

Within this research, we utilized HOMER Pro version 3.14.2 to perform approximately 862,488 simulations within a duration of about 3 hours and 35 minutes. The analysis involved the examination of 5324 optimization scenarios and 81 sensitivity cases. As previously outlined, the primary goal of the investigation was to optimize the dimensions and costs of an integrated energy system to meet the energy requirements of the designated region. Leveraging the available database, we subjected the proposed configuration to simulation using the HOMER Pro tool.

3.1. Results of the optimization process for the system

This study initiates with specified conditions, including a daily load demand of 15545 kWh, an average annual wind speed of 3.88 m/s, solar irradiation averaging 5.01 kWh/m²/day, annual average biomass scale set at 299.51 t/d, a Nominal Discount Rate (NDR) set at 8.9 %, and a

Table 10
Sensitivity inputs.

Sensitivity variables	Values
solar radiation (kWh/m ² /day)	5.01 – 5.5 – 6
wind speed (m/s)	3 – 3.88 – 4.5
biomass availability (t/d)	250 – 299.51 – 350
annual average electric load (kWh/day)	14500 – 15545.13 – 16500

Table 11

The comprehensive characterization of cost-effective configurations for HRES employing a CC strategy.

Sizing	System 1	System 2	System 3	System 4	System 5	System 6
	PV/WT/BMG	PV/BMG	WT/BMG	PV/WT	PV	WT
PV (kW)	3877	6321	0	5267	12684	0
WT (kW)	1650	0	8250	4950	0	16500
BMG (kW)	500	500	500	0	0	0
Batteries (kWh)	5788	9880	14960	16888	28376	67828
Converter (kW)	2224	2483	2600	3040	3147	3820
NPC (M\$)	8.29	10.00	16.20	16.70	21.20	45.20
COE (\$/kWh)	0.125	0.150	0.244	0.251	0.319	0.680
Operating cost (M\$/yr)	0.167	0.212	0.264	0.265	0.385	0.720
Initial capital (M\$)	6.34	7.51	13.1	13.6	16.7	36.8
Excess Elec. (MWh/year)	5218.45	7024.58	7937.73	10027.78	15395.88	17393.99
CO ₂ emission (tonnes/year)	16.341	17.201	15.367	0	0	0

projected lifespan of 20 years. Utilizing HOMER software, six potential configurations for HRES are suggested to meet the power demand of the village. The system's dispatch strategy involves cycle charging (CC). Table 11 outlines the key characteristics of the most economically viable HRES configuration among the six feasible hybrid systems. Notably, Systems 1, 2, and 3 incorporate a biomass generator (BMG) as the backup system, whereas Systems 4, 5, and 6 do not include a biomass generator in the HRES.

From the preceding table, it is evident that System 1 stands out as the optimal configuration. Taking into account the initial parameters, the PV/WT/BMG configuration proves to be the most cost-effective among the architectural choices, boasting a NPC of 8.29 M\$. This setup comprises a 3877 kW photovoltaic array, one 1.65 MW wind turbines, 5788 batteries with a 1 kWh capacity each, and a 2224 kW converter. As delineated in Table 10, the PV/WT/BMG system exhibits a COE of 0.125 \$/kWh, an annual operating cost of 0.167 M\$, and a capital investment of 6.34 M\$.

Following the PV/WT/BMG system, the second most advantageous hybrid setup is the PV/BMG, presenting an NPC, COE, and initial investment of 10 M\$, 0.150 \$/kWh, and 7.51 M\$, respectively. In close succession, the third-ranked hybrid system is the WT/BMG, with corresponding values of 16.2 M\$ for NPC, 0.244 \$/kWh for COE, and 13.1 M\$ for the initial investment. The fourth-ranked hybrid system is the PV/WT, with corresponding values of 16.7 M\$ for NPC, 0.251 \$/kWh for COE, and 13.6 M\$ for the initial investment. The fifth-ranked system is the only PV, with corresponding values of 21.2 M\$ for NPC, 0.319 \$/kWh for COE, and 16.7 M\$ for the initial investment. The last-ranked system is the only WT, with corresponding values of 45.2 M\$ for NPC, 0.680 \$/kWh for COE, and 36.8 M\$ for the initial investment. It is noteworthy that standalone PV or WT systems are considerably more expensive, underscoring the cost-effectiveness of hybrid systems over single-source energy approaches [46].

In comparison to the PV/WT/BMG configuration, all other configurations demonstrate slightly reduced cost-effectiveness. This decline can

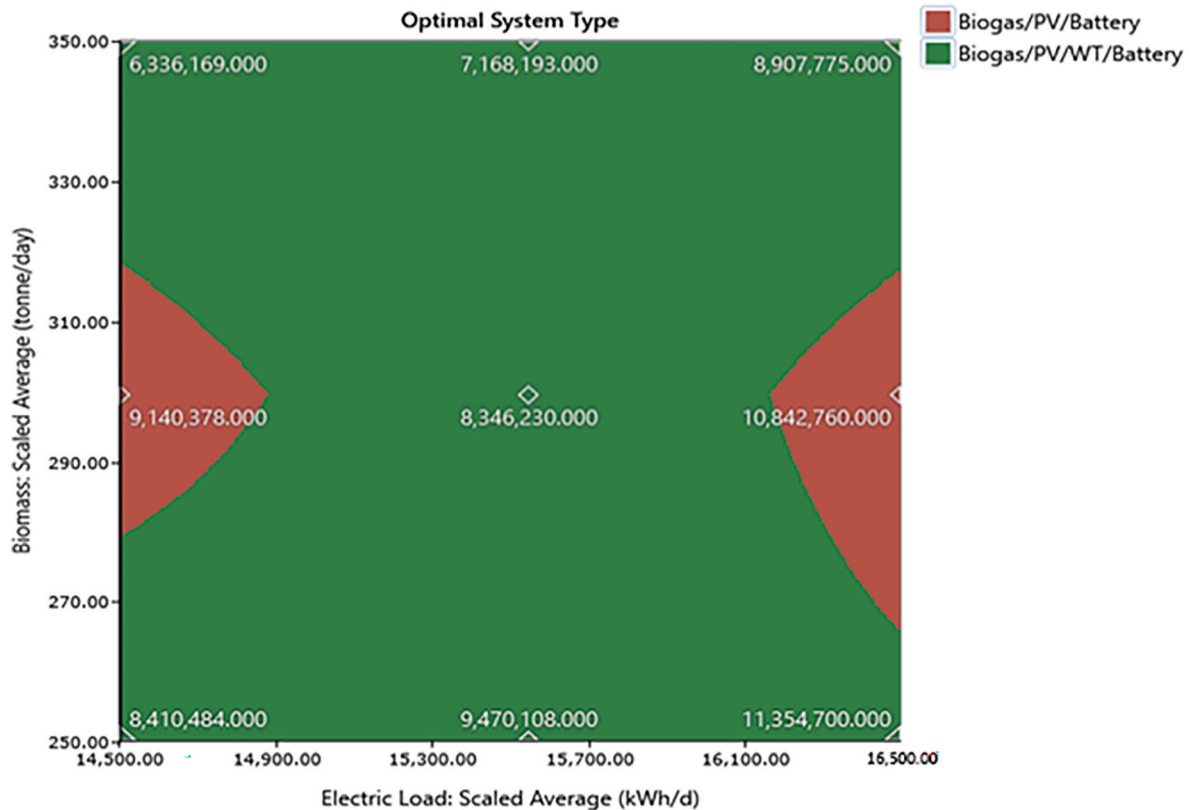


Fig. 15. Determining the optimal system type under various scenarios: Annual Electric Load versus biomass availabilities, with constant solar radiation and wind speed at 5.01 kWh/m²/day and 3.88 m/s, respectively.

be attributed to the higher quantity of batteries in the other configurations, leading to increased costs.

3.2. Sensitivity analysis results

Conducting a sensitivity analysis, we examined various variables outlined in Table 10 to gauge the efficiency of the designed hybrid system under different conditions. The study delved into the impacts of fluctuations in solar irradiation, wind speed, biomass availability, and annual average electric load on determining the most suitable system type. Three scenarios were considered with biomass availabilities set at 250 t/d, 299.51 t/d, and 350 t/d. Additionally, annual average electric load values were set at 14500 kWh, 15545.13 kWh, and 16500 kWh, while keeping solar radiation and wind speed constant at 5.01 kWh/m²/day and 3.88 m/s, respectively. In the majority of cases examined, the PV/WT/BMG/Battery system configuration emerged as the preferred architecture, as depicted in Fig. 15. These findings underscore the adaptability of the optimal design across varying biomass availabilities and annual average electric loads.

Moreover, an examination of the effects of varying biomass availability and annual average electric load on the Net Present Cost (NPC) and Cost of Electricity (COE) for the optimal PV/Wind/BDG/Battery system is detailed in Table 12. With an increase in electric load from 14500 kWh to 16500 kWh and a decrease in biomass availability from 350 t/d to 250 t/d, the COE experiences a rise from 0.102 \$/kWh to 0.161 \$/kWh, accompanied by an NPC increase from 6.34 M\$ to 11.4 M\$. These results highlight the sensitivity of the system's economic metrics to changes in electric load and biomass availability, emphasizing the importance of considering these factors in the overall system design and planning. Furthermore, the economic viability of the PV/WT/BMG/Battery system surpasses that of the traditional system (diesel generator) across all the scenarios examined. In summary, variations in solar irradiation, wind speed, biomass availability, and scaled annual average electric load do not exert a very impact on the economic viability of the hybrid power system in this study.

Wind speed and solar radiation were systematically adjusted to assess their respective impacts on the economic performance of the most efficient system. Surface plots were employed to visualize the effects of varying wind speed and solar irradiation on the NPC and COE of the system. Throughout these simulations, the annual electric load and annual biomass availability were held constant at 15545.13 kWh and 299.51 t/d, respectively. Fig. 16 depicts a surface plot with the NPC of the hybrid system, overlaying COE, where the annual average solar irradiation is represented on the x-axis and wind speed on the y-axis.

The plot reveals that with a rise in solar irradiation from 5.01 kWh/m²/day to 6.00 kWh/m²/day (at a constant wind speed of 3.88 m/s), the COE declines from 0.125 \$/kWh to 0.121 \$/kWh, accompanied by a drop in the NPC from approximately 8.29 M\$ to 8.02 M\$. This is attributed to the augmented net power output from PV panels as solar radiation intensifies, leading to a reduction in NPC. As the overall NPC decreases, there is a corresponding decrease in the COE of the system. Similarly, when wind speed is increased (at a constant solar radiation of

5.5 kWh/m²/day), an escalation from 3.00 m/s to 4.50 m/s results in a decline in COE from 0.144 \$/kWh to 0.099 \$/kWh, alongside a decrease in NPC from about 9.57 M\$ to 6.61 M\$. Thus, an elevation in wind speed contributes to a simultaneous reduction in both COE and NPC. This underscores that higher solar radiation and wind speed positively impact both the cost-effectiveness and economic viability of the hybrid system, leading to lower COE and NPC. Consequently, the hybrid PV/WT/BMG/Battery system proves to be more cost-effective than conventional alternatives, primarily due to its adaptability to natural energy resources. Specifically, in extreme conditions with low solar radiation, wind speed biomass availability and high annual electric load (5.01 kWh/m²/day, 3.00 m/s, 250 t/d and 16500 kWh/d), as depicted in Fig. 17, the COE and NPC of the system are 0.167 \$/kWh and 11.8 M\$. These results highlight the sustained economic viability of the most efficient hybrid system compared to traditional approaches, even under extreme conditions considered in this study.

The availability of renewable energy from wind, solar, and biomass depends directly on natural factors like wind speed, solar radiation, and biomass supply. Decreases in these factors significantly reduce the efficiency of renewable energy systems. Lower wind speeds and reduced solar radiation decrease electricity generation from turbines and solar panels, respectively. Limited biomass availability similarly hinders bioenergy production. These relationships, supported by earlier equations, highlight the critical impact of natural resource variability on renewable energy technology effectiveness. Managing these fluctuations is crucial for ensuring consistent renewable energy output.

3.3. Simulation results for the optimal hybrid system

The analysis conducted above underscores the superiority of the PV/WT/BMG hybrid system as the preferred choice for supplying power to the city. This combined setup yields a total annual electricity output of 11.14 GWh, as illustrated in Figs. 18 and 19. Notably, wind turbines account for 21.1 % of this production, while photovoltaic (PV) panels contribute the majority at 59.5 %, and the Biogas generator contributes 19.3 % to the overall production. Figs. 20 and 21 showcase the daily electricity generation of the system during winter (31 December) and the summer (14 August) months, respectively.

As depicted in Fig. 20, wind generators and solar panels effectively meet the electrical load demands for a significant portion of the day. Yet, from 12:00 a.m. to 9:00 a.m., the shortfall in solar and wind energy—potentially due to calm conditions at night or insufficient/excessive wind speeds—creates a gap in fulfilling energy requirements. During this interval, the biogas generator steps in to generate energy, maintaining balance. Between 5:00 AM and 9:00 AM, the batteries come into play, discharging stored energy to support the biogas generator.

Following this, solar panels, wind turbines, and biomass generator resume energy generation, satisfying electricity demands and storing excess energy in batteries, the biomass generator continues operation until 11 a.m. when the batteries reach full charge, at which point the biomass generator is deactivated. However, between 5:00 p.m. and 11:00 p.m., there is a reduction in solar wind energy output, and energy demand sharply rises, promptly met by the BMG and battery reserves. These batteries also facilitate the reactivation of wind turbines, compensating for reduced operation during periods of lower wind speeds. Furthermore, the absence of sunlight from 6:00 p.m. to 8:00 a.m. results in no nighttime solar power generation.

In contrast to the winter conditions, as depicted in Fig. 21, wind energy is entirely absent in this scenario, attributed to the lack of wind speed. However, the timeframe between 7 a.m. and 7 p.m. solar panels recommence energy generation, satisfying electricity demands and storing excess energy in batteries, eliminating the need for supplementary power sources. Nevertheless, after 7 p.m., there is a significant increase in energy demand, prompting the discharge of battery reserves and the generation of energy from biogas until 8 p.m., at which point the energy demand and battery discharging gradually diminish.

Table 12
Effects of biomass availability and electric load on the system.

System design	biomass availability (t/d)	electric load (kWh/d)	NPC (M\$)	COE (\$/kWh)
PV/WT/BMG/Battery	350	14500	6.34	0.102
	350	15545.13	7.17	0.108
	299.51	15545.13	8.29	0.125
	250	14500	8.41	0.136
	350	16500	8.91	0.126
	299.51	14500	9.14	0.147
	250	15545.13	9.47	0.142
	299.51	16500	10.8	0.154
	250	16500	11.4	0.161

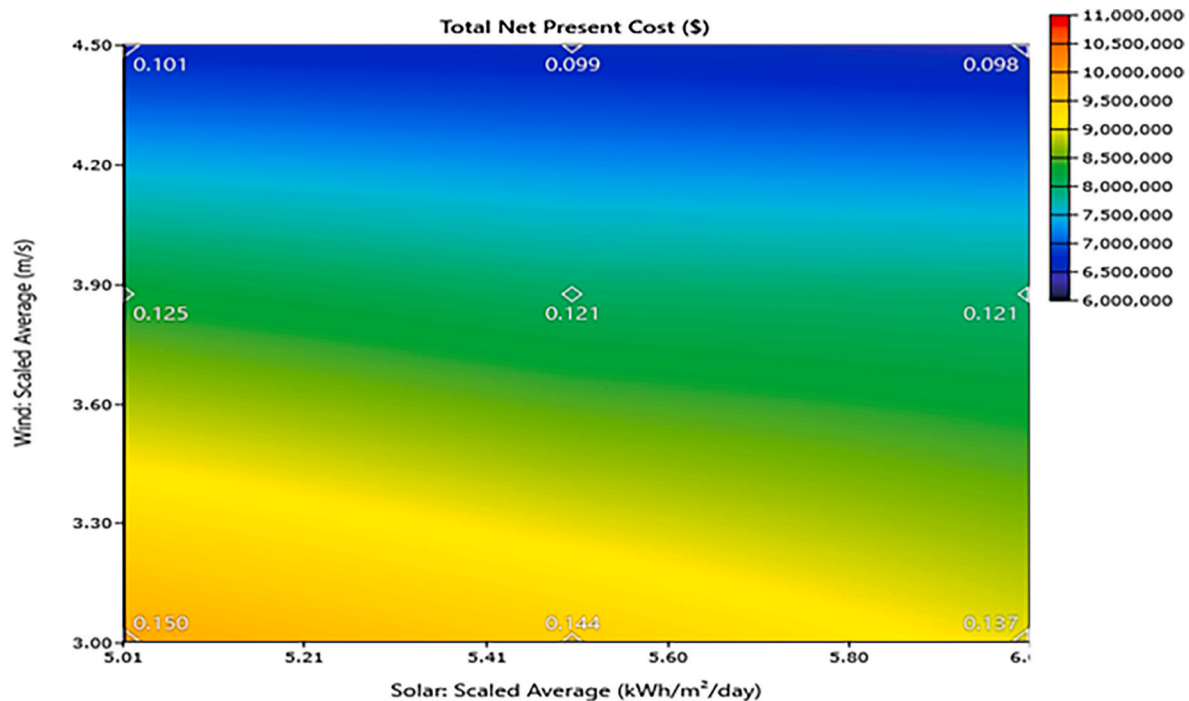


Fig. 16. Solar radiation vs Wind speed, the annual electric load and annual biomass availability were held constant at 15545.13 kWh/d and 299.51 t/d, respectively.

Due to the absence of sunlight, there is no solar energy generation from 7 p.m. until 7 a.m. the following day. Nevertheless, this phase becomes advantageous for the biomass generator, which can still contribute to maintaining energy balance during these hours.

As shown in Fig. 22, the photovoltaic (PV) system contributes a total annual electricity production of 6.63 GWh, with operational hours totaling 4383 hours per year and a COE of 0.0355 \$/kWh. Simultaneously, the wind turbine generates 2.36 GWh of electricity annually, operating for 6594 hours per year and having a COE of 0.0633 \$/kWh. Additionally, the biomass generator produces 2.15 GWh of electricity yearly, running for 4509 hours annually, consuming approximately 90689 tons per year, and having a lifespan of 4.44 years. The battery system maintains a 8-hour autonomy, receiving 974.13 MWh of energy annually and delivering 877.66 MWh each year. With a lifespan of 15 years, this battery system incurs total electrical losses of 97.46 MWh per year. The converter remains operational for a significant duration, approximately 5111 hours per year, primarily in AC mode to serve the majority of the loads.

Fig. 23 displays the cost breakdown for the various components of the system, categorized accordingly. Notably, solar panels incur the highest initial capital expense, followed by batteries, wind turbines, system converters, and the biomass generator. Operating costs are primarily attributed to the PV system, followed by wind turbines, then batteries, and finally the BMG. Regarding replacement costs, batteries require the most substantial investment, followed by the biomass generator, and then the converters. The data depicted in Fig. 24 indicates that both battery and converter system replacements are necessary in the 15th year. Throughout the project's lifespan, the PV system requires nearly annual operation and maintenance.

From an economic standpoint, the PV/WT/BMG configuration emerges as the most financially efficient choice, boasting a minimal NPC of 8.29 M\$ and a favorable energy cost of 0.125 \$/kWh. This research assesses the environmental advantages of the PV/WT/BMG system by measuring the decrease in CO₂ emissions in comparison to a diesel generator as an energy source. To sustain the community's electricity needs, the diesel generator must generate a total of 5671.80 MWh/year. As a result, the CO₂ emissions from the electricity generated by the diesel generator amount to 5910.67 tons/year, considering the quantity

of CO₂ emitted per liter of fuel consumed by the diesel generator varies based on the specific attributes of both the generator and the fuel. Typically, this emission factor falls within the range of 2.4–2.8 kg/l [47]. Clearly, the optimized system demonstrates a substantial reduction in CO₂ emissions throughout the project's lifespan, with the proposed system having the potential to avert around 5894.33 tons/year of CO₂ emissions compared with the diesel generator. Furthermore, by achieving significant CO₂ reduction and employing resources wisely, the PV/WT/BMG system not only helps to mitigating greenhouse gas emissions and preserving the environment but also improves local living conditions and overall population health to some extent.

From an energy perspective, the implementation of the PV/WT/BMG hybrid system ensures a reliable and sustainable energy supply, significantly reducing dependency on non-renewable energy sources such as diesel. The integration of solar, wind, and biomass resources guarantees a diversified energy mix that enhances the stability and resilience of the local energy infrastructure. This system also addresses the specific issue of electricity outages in Zoumi's circle, especially during the summer months when energy demand spikes due to increased agricultural activities. By utilizing a hybrid approach, the energy supply becomes more consistent and capable of meeting peak demands efficiently.

Economically, the low cost of energy at 0.125 \$/kWh, coupled with government incentives and subsidies, makes the PV/WT/BMG system an attractive investment. This economic feasibility is crucial for encouraging the adoption of renewable energy systems in rural areas. The reduced net present cost (NPC) of 8.29 M\$ highlights the long-term financial benefits of transitioning to a hybrid renewable energy system, offering a cost-effective alternative to conventional energy sources.

Environmentally, the PV/WT/BMG system offers significant benefits by drastically reducing CO₂ emissions. The reduction of approximately 5894.33 tons/year of CO₂ emissions contributes to Morocco's environmental goals and aligns with global efforts to combat climate change. This substantial decrease in greenhouse gas emissions not only helps in mitigating climate impact but also contributes to cleaner air and healthier living conditions for the local population. The environmental sustainability of the PV/WT/BMG system supports broader ecological preservation efforts and promotes a more sustainable future.

Furthermore, Morocco demonstrates substantial backing for

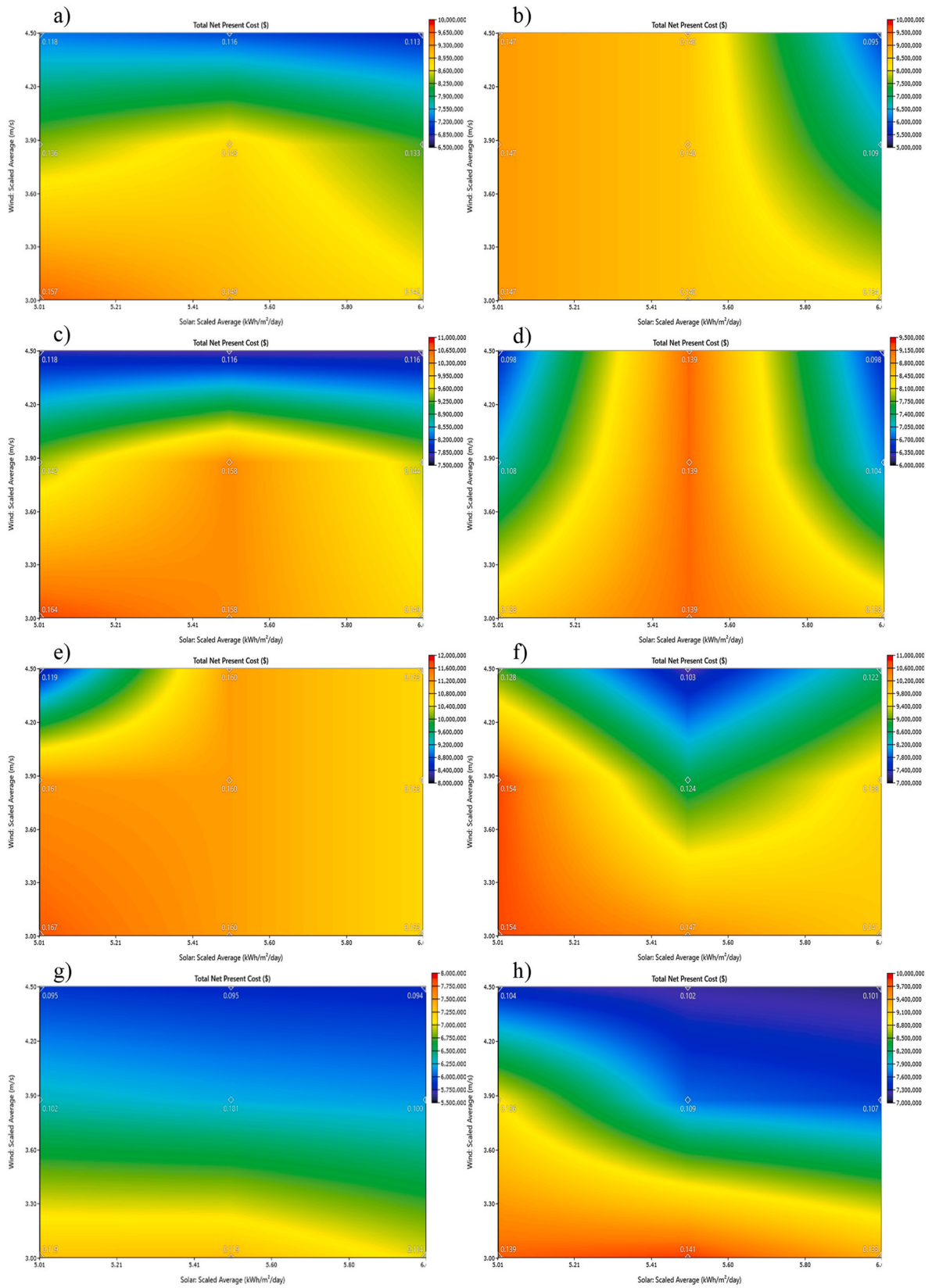


Fig. 17. Solar radiation vs Wind speed, the annual electric load is 14500 kWh/d for (a, b, g) and 16500 kWh/d for (e, f, h) with annual biomass availability ranging from 250 to 350 t/d while the annual electric load is 15545.13 kWh/d for (c, d) with annual biomass availability = 250, 350 t/d, respectively.

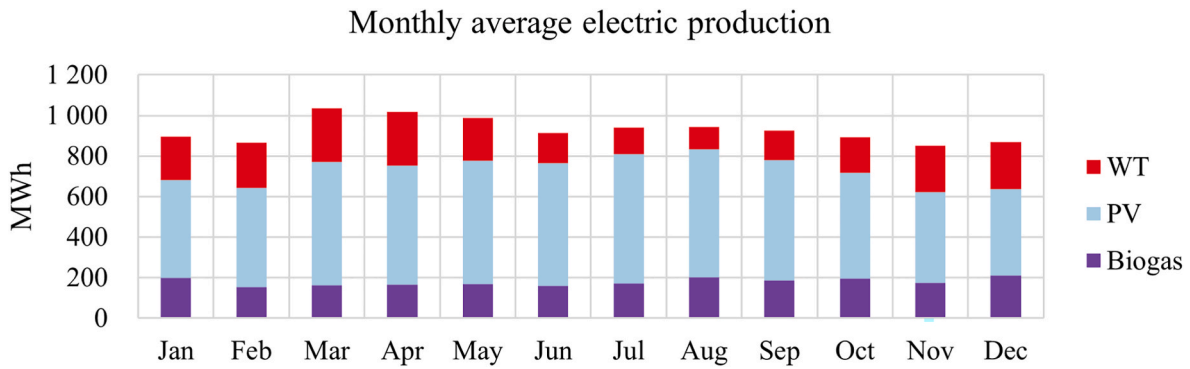


Fig. 18. Average monthly electricity output of the PV/WT/BMG hybrid system.

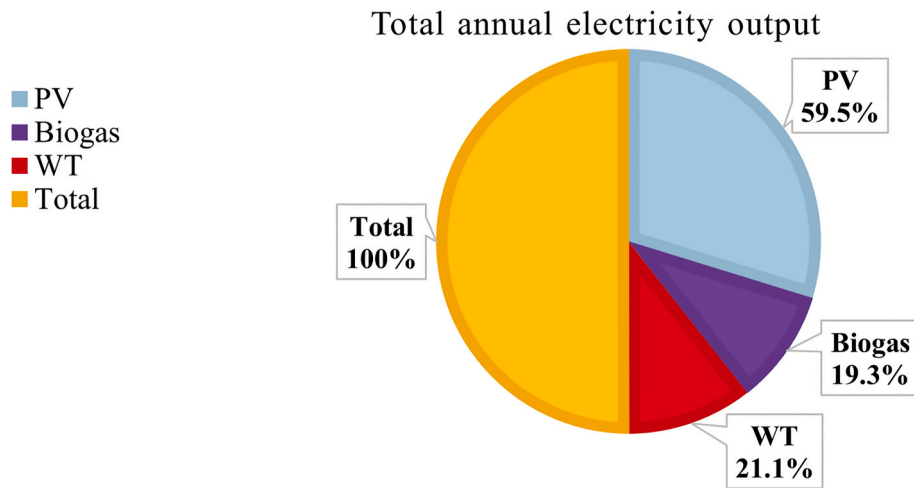


Fig. 19. Monthly electricity generation during a year.

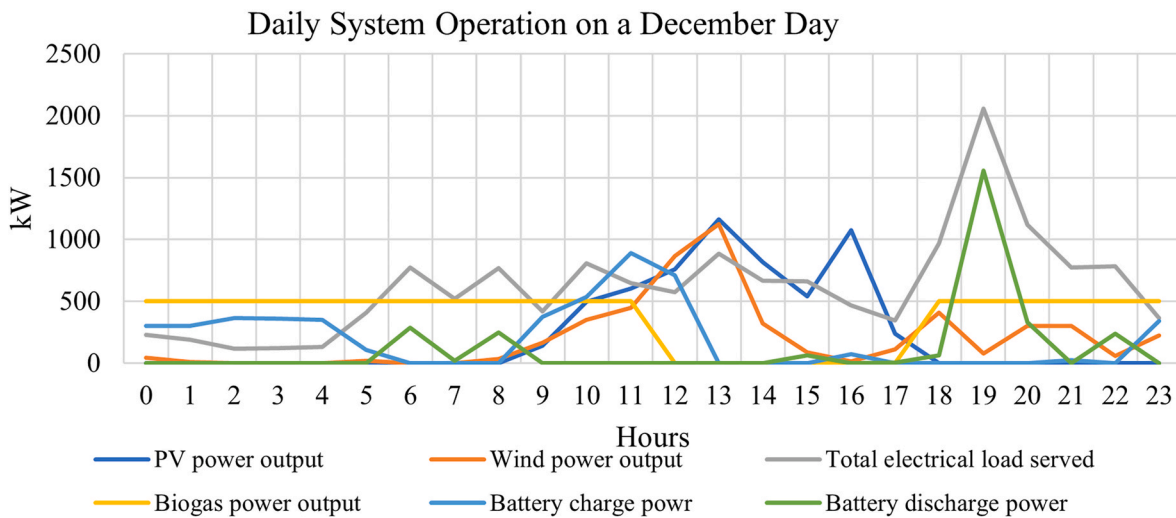


Fig. 20. Daily system performance on a December day.

renewable, sustainable energy sources. Considering the available incentive mechanisms and supportive policies facilitated through government subsidies, potential customers could witness feasible reductions in the investment and equipment expenses of the proposed hybrid system. Morocco, having entered into various agreements such as the Morocco Agreement and partnering with the International Renewable Energy Agency, has actively participated in initiatives like supporting

the cooperative framework on green hydrogen. This collaboration, undertaken in conjunction with the European Commission, addresses the challenges associated with developing infrastructure, technologies, and credits essential for establishing a global green economy and facilitating the trade of green hydrogen [48].

Moreover, the sustainability of the proposed hybrid power system extends beyond its environmental aspects to encompass social and

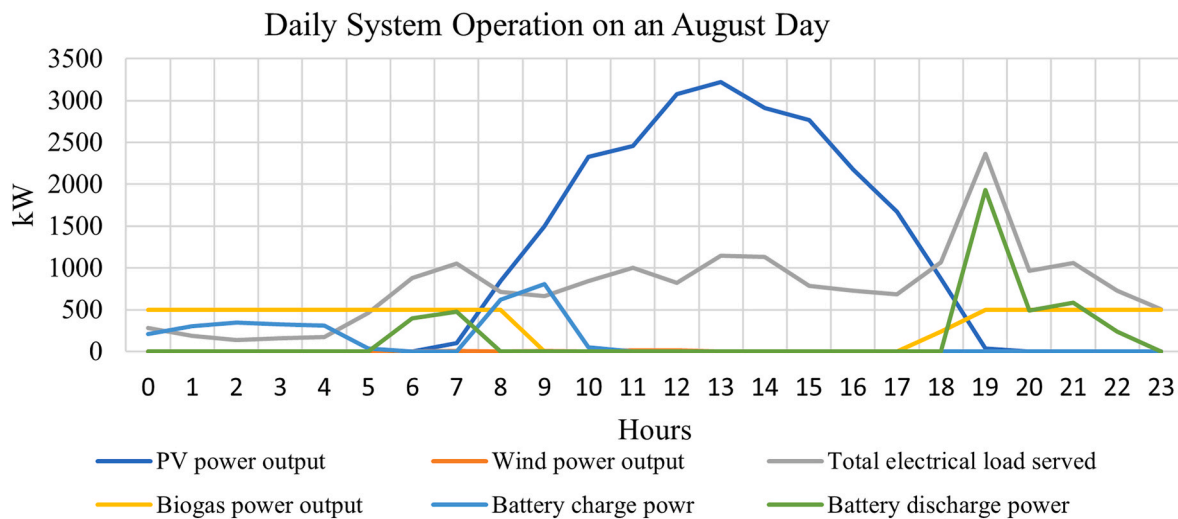


Fig. 21. Daily functioning of the system on an August day.

cultural dimensions. Electricity plays a crucial role in improving the overall quality of life in communities, particularly in remote areas. Considering that a substantial portion of the local population is occupied during daylight hours, bringing electricity to this region can engage people in various activities and enrich their lives after sunset. The establishment of a reliable and sufficient power supply allows individuals to enjoy entertainment through TV and radio, thereby staying connected with news. Additionally, the installation of streetlights contributes to enhanced safety during nighttime movements. From an educational perspective, students benefit from extended study hours facilitated by improved lighting quality at night, significantly influencing the development of local culture [49]. Furthermore, the ongoing need for system operation, maintenance, and potential future upgrades creates new employment opportunities for local residents, fostering sustainable social progress in the long term [50].

4. Conclusion

Developing a sustainable future energy system is paramount to addressing the energy crisis and reducing greenhouse gas emissions in developing nations. Despite existing research, a knowledge gap persists in the study of hybrid power systems, particularly those involving wind, solar, and biomass components, for electrifying rural areas in developing countries. This research focuses on evaluating the viability of a hybrid energy system utilizing solar, wind, and biomass sources to produce electricity for the Zoumi circle in Morocco.

Among the six scenarios explored—PV/WT/BMG, PV/BMG, WT/BMG, PV/WT, only PV, and only WT—the PV/WT/BMG/Battery configuration emerges as the most cost-effective, achieving a high Renewable Energy (RE) fraction of 100 %. The optimized system, designed in HOMER Pro, includes 3877 kW of PV modules, a 1.65 MW wind turbine, a 500-kW biogas generator, 5788 kWh of storage batteries, and a 2224 kW converter. This configuration can produce approximately 11.14 GWh/year, effectively meeting the electricity demands of the Zoumi circle and allowing surplus electricity to be stored in batteries. The electricity is provided to consumers at an affordable rate of 0.125 \$/kWh, and the NPC for this system amounts to 8.29 M\$.

From an economic standpoint, the PV/WT/BMG configuration emerges as the most financially efficient choice, boasting a minimal NPC of 8.29 M\$ and a favorable energy cost of 0.125 \$/kWh. Furthermore, Morocco demonstrates substantial backing for renewable, sustainable energy sources. Considering the available incentive mechanisms and supportive policies facilitated through government subsidies, potential customers could witness feasible reductions in the investment and

equipment expenses of the proposed hybrid system.

This study also assesses the environmental advantages of the PV/WT/BMG system by measuring the decrease in CO₂ emissions in comparison to a diesel generator as an energy source. The system demonstrates a substantial reduction in CO₂ emissions, with the proposed system having the potential to avert around 5894.33 tons/year of CO₂ emissions compared with the diesel generator. By achieving significant CO₂ reduction and employing resources wisely, the PV/WT/BMG system not only helps mitigate greenhouse gas emissions and preserve the environment but also improves local living conditions and overall population health to some extent.

To gauge uncertainty, a sensitivity analysis explores the impact of solar irradiation, wind speed, biomass availability, and scaled annual average electric load on determining the optimal system type and their influence on electricity prices. The analysis highlights those variations in these factors do not significantly impact the economic viability of the hybrid power system, underscoring the system's robustness.

The study's outcomes underscore that a reliable electricity supply through hybrid systems can enhance the quality of life and contribute to ecological and economic growth in Zoumi's circle. As both rural and urban areas in Morocco continue to develop, the evolving consumption patterns and changing power requirements for various user categories will influence the optimal configuration of hybrid energy systems, an aspect worth exploring in future research endeavors. Furthermore, this research lays the groundwork for future studies to investigate the scalability of such systems in other regions with similar conditions and the potential for integrating additional renewable energy sources to further enhance system efficiency and sustainability.

This research provides a comprehensive evaluation of a hybrid energy system combining solar, wind, and biomass sources. The findings suggest that Zoumi is a suitable location for electricity generation using a hybrid wind-solar-biomass mix. The evaluation of resources and the feasibility study will assist the local government in attracting potential investors, whether local or international, to fund the project, ultimately contributing to the sustainable development of the region.

CRedit authorship contribution statement

Abdellah El-Maaroufi: Writing – original draft, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Mohammed Daoudi:** Writing – review & editing, Supervision. **Rachid Ahl Laamara:** Writing – review & editing, Supervision.

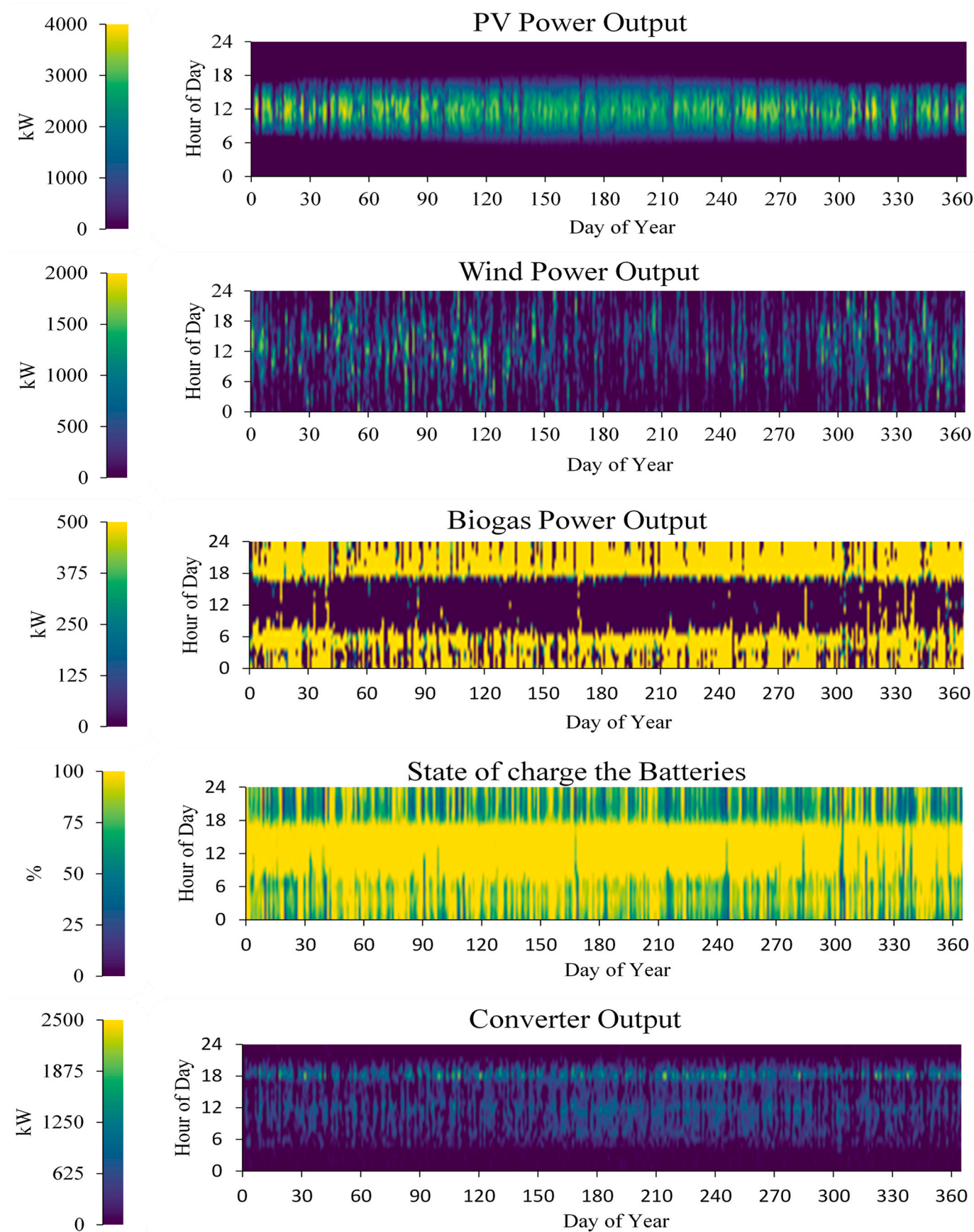


Fig. 22. Simulation results for each component.

The NPC for each individual component

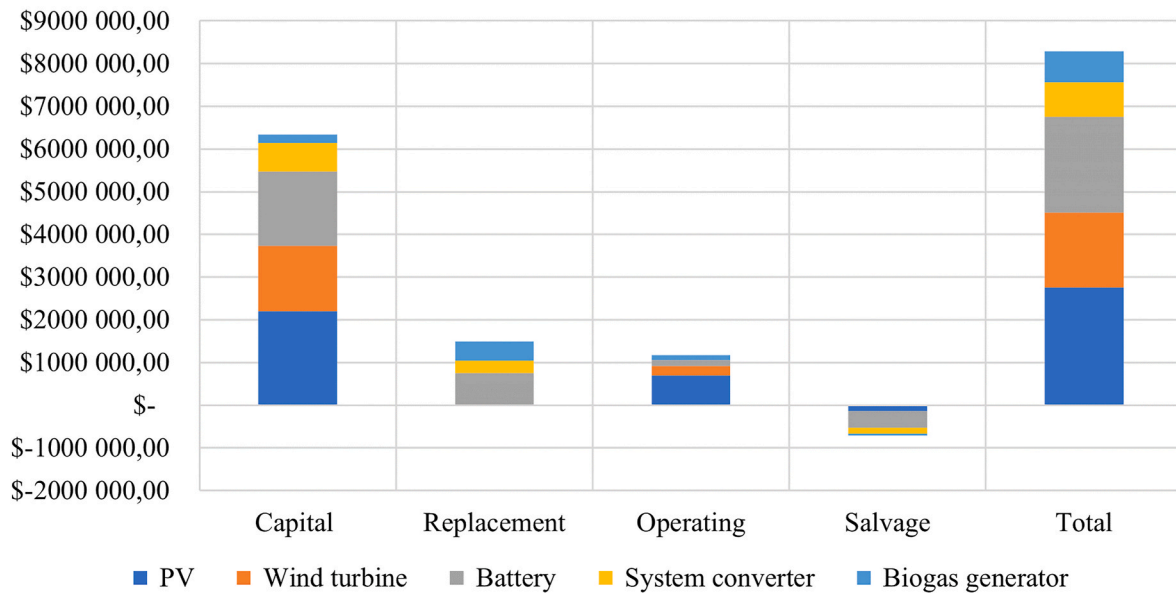


Fig. 23. Cost breakdown of PV/WT/BMG hybrid System by category.

Cash flow by cost type

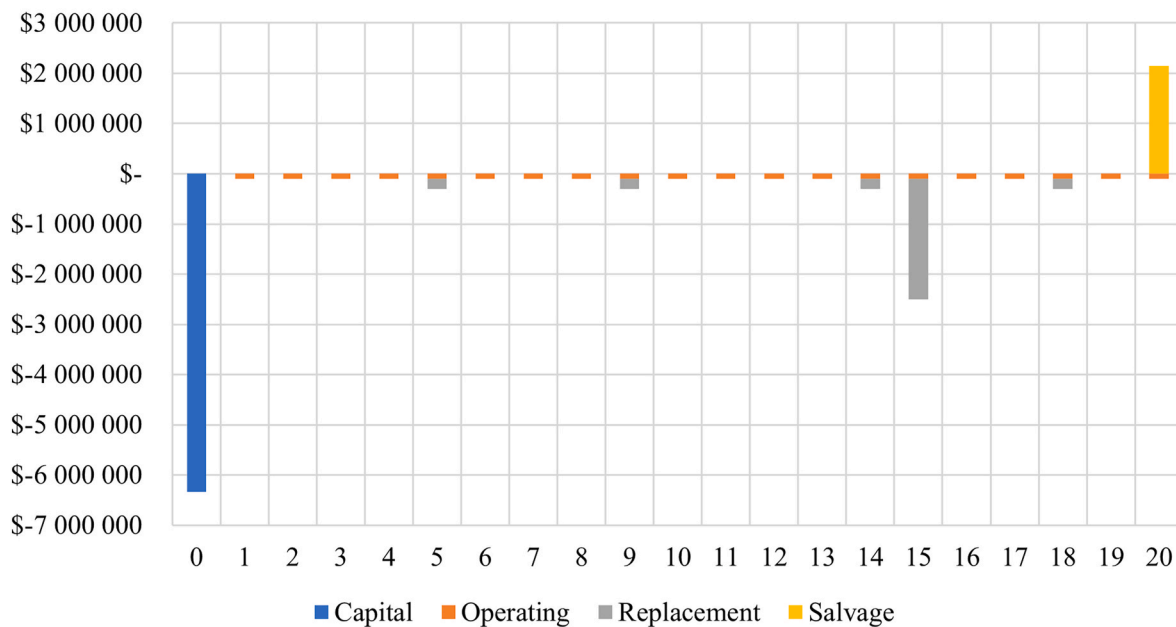


Fig. 24. PV/WT/BMG system—Cash flow.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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